

ABSTRACT

Distribution and Biology of the Spider Crab Chionoecetes opilio in the Magdalen Shallows, Gulf of St. Lawrence.

Howard Powles.

Submitted for the M.Sc. degree in the Department of Marine Sciences.

Chionoecetes opilio is concentrated in troughs at the north-western and southeastern ends of the Magdalen Shallows. Bottoms of the troughs are in the cold intermediate layer, and their predominant bottom sediment is mud. C. opilio is most abundant on mud bottoms and at temperatures of -1°C to $+2^{\circ}\text{C}$. The crab is inactive, buried in the bottom mud, during daylight hours. Growth of the males in the final moult is estimated at 38% of width, 130% of weight. The estimated 50% maturity size for both sexes is 50 mm. carapace width. No newly-moulted mature females were found, and since the males grow larger than females it was concluded that females cease growing at maturity. Eggs are laid in late May and are carried by the females until hatched in the following April. The crab is subarctic in the northwest Atlantic, and is found in winter-cooled waters of the northwest Pacific.

Short title:

Chionoecetes opilio in the Magdalen Shallows

Distribution and Biology of the Spider Crab
Chionoecetes opilio in the Magdalen Shallows,
Gulf of St. Lawrence

by

Howard Powles

Thesis submitted in partial fulfillment of the
requirements for the M.Sc. degree in the
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Description of the species

Chionoecetes opilio belongs to the family Majidae (the spider crabs), subfamily Pisinae. The carapace is about as broad as long, with the branchial regions far apart. The two-toothed rostrum is horizontal, and the orbits are large, lacking a pre-orbital tooth but with a prominent post-orbital tooth. The gastric region of the carapace is depressed, and separated from the branchial regions by a ridge on each side extending posteriorly from the post-orbital tooth regions. The two ridges curve inward and meet in the cardiac region. The carapace is partly covered with tubercles, which take two forms, one granular, blunt and wartlike, the other sharp and smooth. The former type occurs scattered over the carapace, and in a transverse row through the cardiac region, behind which no tubercles occur. The sharper denticles occur in two rows on each side of the carapace, anteriorly, one following the ridges from post-orbital tooth to cardiac regions, the other more marginal, starting at the angle of the buccal cavity and continuing posteriorly. The legs are considerably longer than the carapace, accounting for the name of the spider crab group. In the male the chelipeds are 1.5 to 2 times the total length of the carapace (rostrum included); the meropodite has a row of denticles on its anterior edge, while two rows of blunt tubercles are found on the manus of the claw, which is roughly cylindrical. The first three pairs of walking legs are 2 to 3 times as long as the carapace in the male, and the segments are dilated, oval in section. The last pair of walking legs is somewhat smaller, about as long as the chelipeds, and round in section. The abdomen has seven segments, the last of which is invaginated into the sixth. Colour varies from dull brick-red to yellowish-white dorsally, yellowish-brown to white ventrally, for both carapace and legs. Sexual dimorphism is

apparent, particularly in the ratios of leg length to carapace length. In the female, the first three pairs of walking legs are 1.5 to 2 times, and the chelipeds and last walking legs 1 to 1.5 times, the carapace length. The female abdomen is wider than that of the male, as an adaptation for carrying eggs. The maximum size of females caught in the Gulf of St. Lawrence is in the neighborhood of 95 mm. carapace width, while the males' size range extends to about 150 mm. These differences in form are illustrated in figure 1.

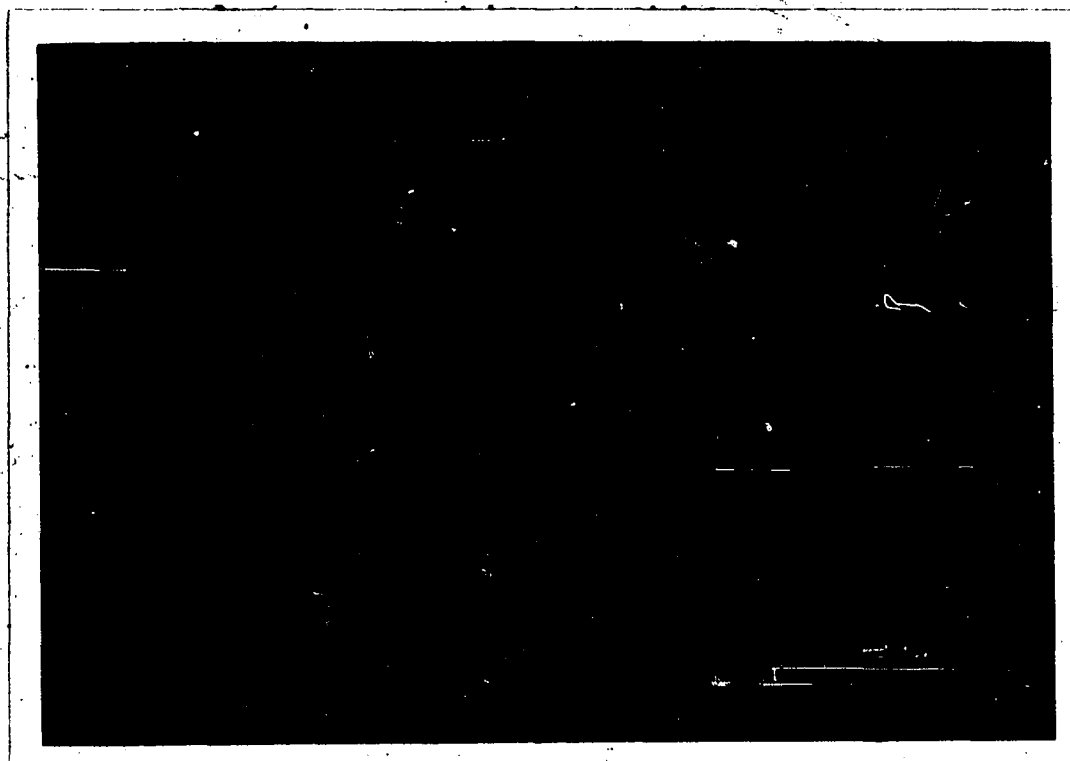


Figure 1. Male (above) and female C. opilio , ventral view.

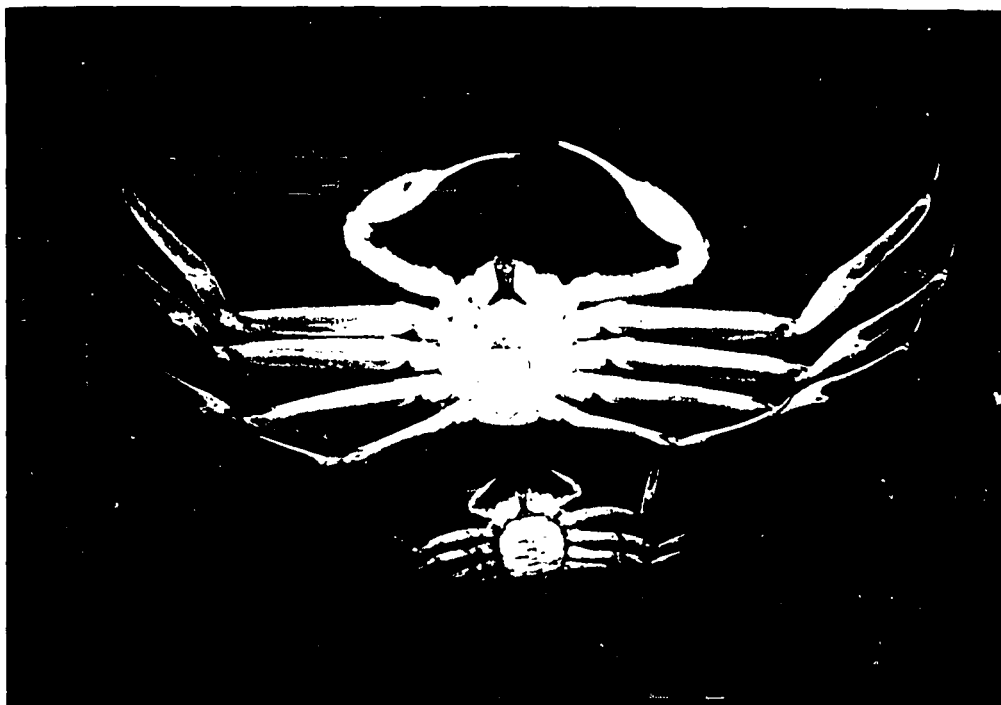


Figure 1. Male (above) and female C. opilio , ventral view.

Literature

Mentions of C. opilio in the literature are few and scattered, most of its occurrences in English works being incidental to a larger investigation. Rathbun (1925) describes the species and summarises its worldwide distribution. More recently, Nesis (1963) summarises the distribution of number of Pacific-West Atlantic (amphiboreal) species, of which C. opilio is one, and describes their zoogeographic history before and since Pleistocene times. Off northwest North America, C. opilio is mentioned by MacGillivray (1955) in a description of the fauna of Point Barrow, Alaska, while American fisheries biologists, Ellson et al., (1950) refer to C. opilio and the closely related C. tanneri in connection with experimental trawling for the King crab (Paralithodes camtschitica) in the Bering Sea. Scholander et al. (1953) used C. opilio (with another Majid crab, Hyas coarctatus) in a study of metabolic adaptation in arctic and tropical animals. Japanese and Russian workers have investigated aspects of the ecology of C. opilio in the Sea of Japan and Kamschatka area; these works are largely in Japanese and Russian, although English abstracts are sometimes present. Ito (1967) writes on reproductive characteristics of the female; Yasuda (1967) gives species composition of stomach contents of the crabs; while Zarenkov (1960) describes aspects of the ecology of decapods of the Kamschatka region, including Chionoecetes opilio. The larvae have been described by Aikawa (1937) and Kurata (1963), who raised eggs and larvae from females from the Sea of Japan; Stephensen (1935) describes a zoea collected off West Greenland which he attributes to C. opilio, but it differs markedly from that of the Japanese writers, who, since they had the eggs from C. opilio females, are to be favored.

In the northwest Atlantic area, Stephensen (1935) gives location and depth of C. opilio finds off West Greenland. Squires (1965) gives a distribution map for C. opilio covering West Greenland, the Canadian east Arctic, and shelf areas from Labrador to northern Maine; this work is recapitulated in his Folio of the Serial Atlas of the Marine Environment (1966). In addition, he briefly describes (1965) size ranges, stomach contents, associated species and female size at maturity. Brunel (1960, 1961, 1962) describes a population of C. opilio off the Gaspé coast in the Gulf of St. Lawrence in terms of catch with time of day and year, depth, bottom type and bottom temperature; size ranges and 50% maturity size of females; and the critical times of moulting, hatching of eggs, and copulation for some parts of the population. Recently, C. opilio in the Gulf of St. Lawrence has been the subject of considerable commercial interest, and several reports on surveys, fishing techniques, and the animal's biology have appeared-- Deveau and Aucoin (1966), Judson and Judson (1966), Wilder (1966), Deveau and McAllister (1967), and Powles (1966, 1968). McLeese (in press) has investigated temperature resistance, and McLeese and Watson (in press) the oxygen consumption, of the crab. Studies of C. opilio are expected to become more extensive with the expansion of the fishery, and literature on the crab in the Canadian Atlantic should expand considerably over the next several years.

Method and Materials

Stations for two cruises on the F.R.B. vessel Harengus in August and September 1967, and the two general areas in which fisheries for C. opilio were concentrated in 1966 and 1967 are indicated in figure 2. The September cruise, off Cape Breton, although of a week's duration, was interrupted by heavy winds and only three tows could be made. Sampling times and places are summarised in Table 1. In addition, data were provided by Groundfish Investigation, F.R.B. St. Andrews, for C. opilio caught over the entire Magdalen Shallows between 1962 and 1966; these were obtained with #35 and #42 otter trawls, with 12" and 16" rollers, respectively, on the foot-ropes, usually towed for half an hour. A #35 otter trawl with 12" rollers was used in the crab fishery off Grande-Rivière in June 1966, tows being of varying duration. A #35 otter trawl was used on the Harengus trips, with two modifications--4" spacers were used on the foot-rope instead of the 12" rollers, and the cod end was lined with 1" stretched mesh length nylon netting--with the intention of putting the foot-rope closer to the ground than in the standard gear, and of retaining small animals which might slip through the standard 4" trawl netting. Trawling exclusively at night was planned for the Harengus trips, since night catches of C. opilio are larger than day catches, but this was not feasible, so most sampling was done during daylight with several night sets for comparison.

Traps used in the fishery were of two general types (see figure 3). Both were rectangular, with two fishing heads at opposite sides, and had a framework made of iron bars of from $\frac{3}{8}$ " to 1" in diameter. The trap used in 1966 off Grande-Rivière measured 5' x $2\frac{1}{2}$ ' x 2',

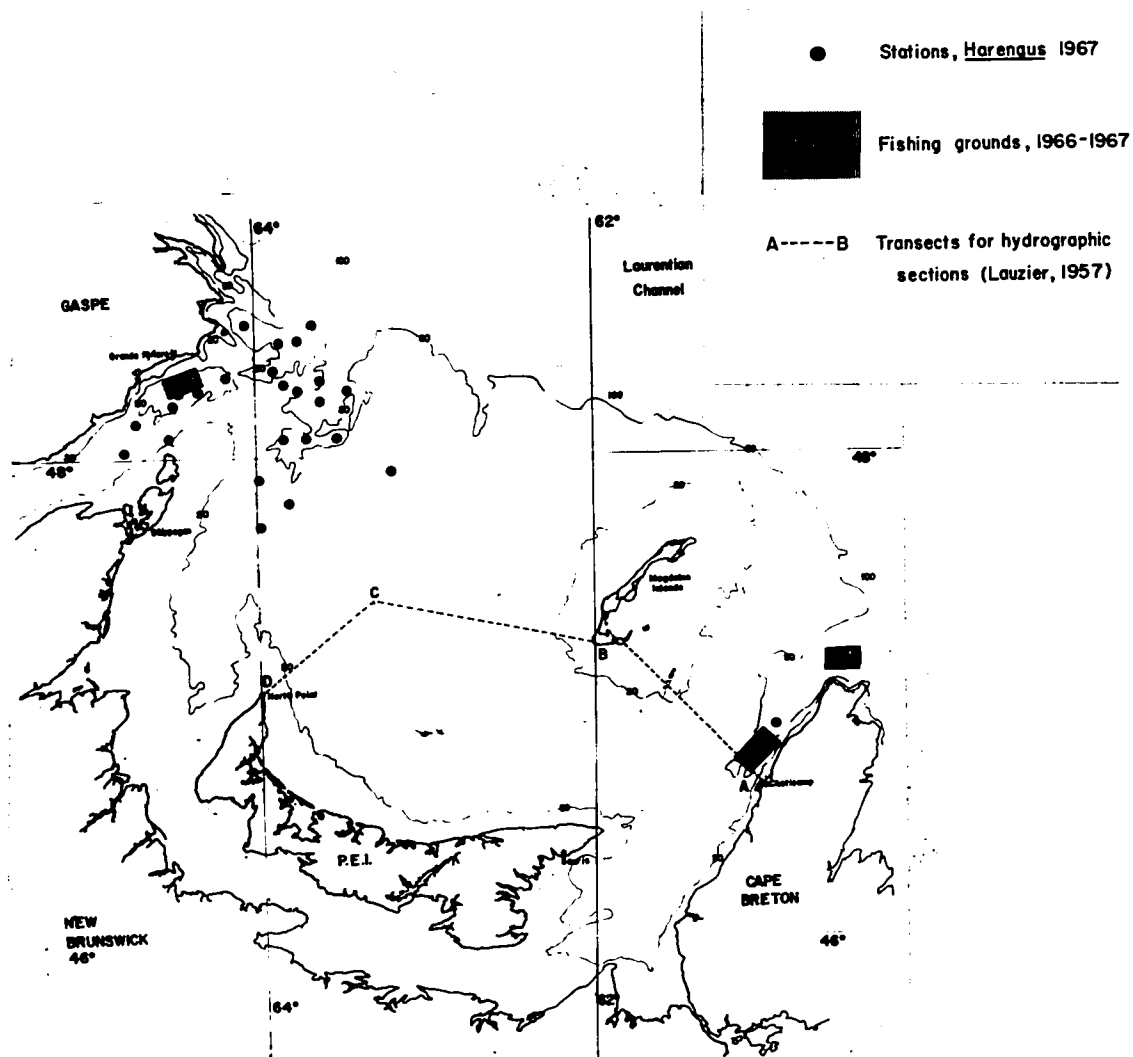


Figure 2. Stations studied on 1967 cruises, grounds fished in 1966 and 1967, and hydrographic sections shown in Figures 6 and 7.

<u>AREA</u>	<u>TIME</u>	<u>GEAR</u>	<u>PURPOSE</u>
A. Magdalen Shallows	1962-1966	Unmodified otter trawl	Groundfish surveys
B. Gaspé coast	June 1967	large trap small mesh	Commercial fishing trials
	July 1966	Unmodified otter trawl	" " "
	August 1966	Small trap	" " "
	August 1967	Modified otter trawl	FRB spider crab survey
C. West Cape Breton	July 1967	large trap medium mesh	Commercial fishing
	September 1-7, 1967	modified otter trawl	FRB spider crab survey
	November 28- December 7, 1967	large trap large mesh	Commercial fishing

Table 1. Cruise times and purposes and sampling methods for C. opilio samples.

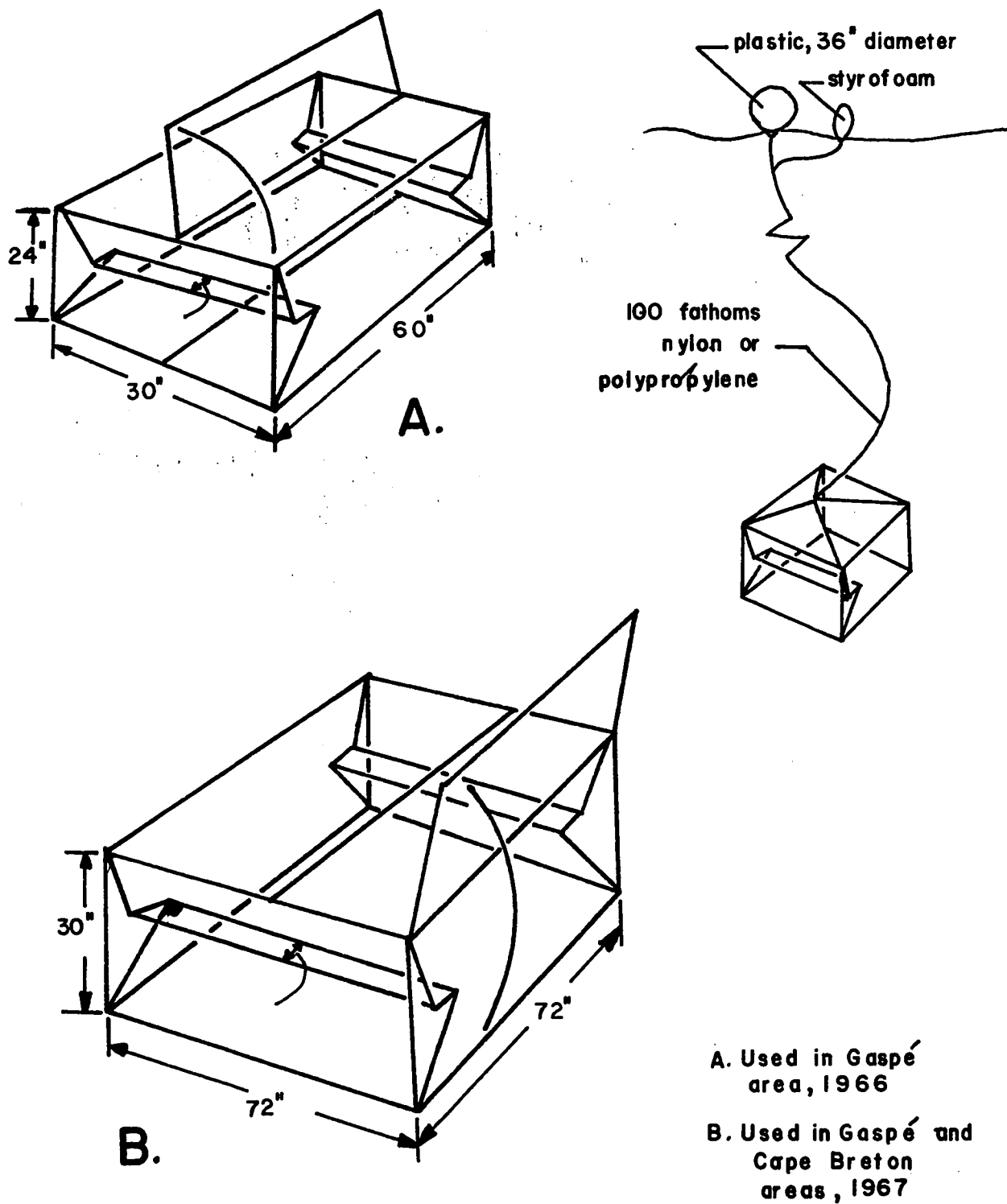


Figure 3. Traps used in fishery for *C. opilio*.

weighed about 100 pounds, and was covered with plastic-coated steel screening, mesh size 1" x 2". Traps used in the Cape Breton area in 1966, and in both areas in 1967, were 6' x 6' x 2½' and weighed 300 pounds. The plastic-coated steel screen was used to mesh in the traps set off Grande-Riviere, while those in the Cape Breton area had polypropylene netting, of stretched mesh length varying from 4" to 6", covering the framework. All traps had 100 fathom lengths of polypropylene or nylon rope attached to a bridle on the trap; at the other end of the rope was a 30" or 36" inflated plastic buoy and a styrofoam trailer buoy to facilitate picking up the trap rope from the vessel. Traps were baited with fresh herring, mackerel or groundfish, or frozen mackerel.

Various measures of C. opilio size are shown in figure 4. Squires (1965) used carapace length (A) from the posterior margin of the eye socket to the most posterior part of the carapace as a general measure of size, while Brunel (1960-1962) used total carapace length (B) from rostrum tip to posterior margin of the carapace. It was felt by the writer that both these size measures had disadvantages. Squires' method involves measuring the distance between a fixed point (the posterior margin of the orbit) and a point in space (point a, directly posterior to the orbit and opposite the posteriormost point of the carapace), instead of between two fixed points. While the error introduced by this method would be effectively negligible (perhaps 2-3 mm), measuring from fixed point to fixed point is both easier and quicker. Breakage of the rostrum would affect Brunel's carapace length measurement, also by requiring use of an estimated point in space. The writer used maximum carapace width ((C)), which is easily measured between two solid points, and is unaffected by light damage to the carapace. Factors for the interconversion of these three measurements are given in table 2. Carapace width of sampled crabs was measured to the nearest millimeter with 210 mm. stainless steel calipers.

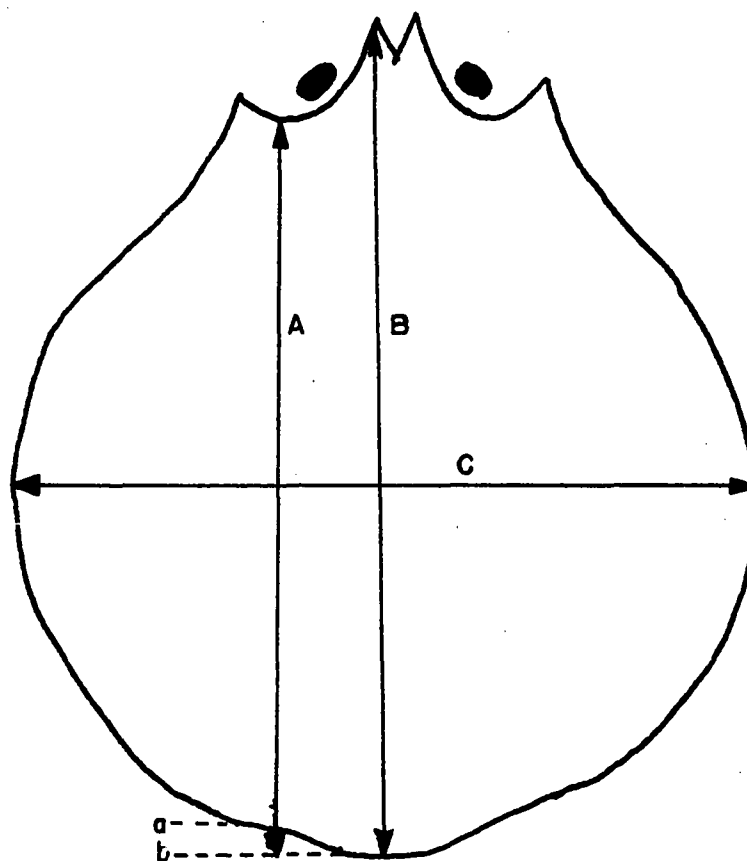


Figure 4. Size measures used in several works on C. opilio.

A. Squires (1965).

B. Brunel (1960, 1961, 1962).

C. Present paper.

x	Number measured	<u>Carapace width</u> x	<u>x</u> Carapace width
Squires' carapace length (A in figure 3)	114	1.20	0.83
Brunel's carapace length (B in figure 3)	203	1.02	0.98

Table 2. Interconversion Factors of carapace dimensions.

The sex of sampled crabs was noted, as was the presence of eggs on the female. Carapace condition was noted as hard or soft; walking leg segments of soft-shelled crabs can be easily bent and broken, while those of the hard-shelled animals resist bending. Carapace condition of hard-shelled animals can be described further by presence of epibiotics (chiefly bryozoans) and colour (which varies from white or pink in the soft-shelled animal to yellow and brown in the old carapace); these characteristics have not been correlated with actual carapace age in weeks or months, and were not used to describe freshly caught crabs. Presence of epibiotics was, however, used as an indication of old carapace in working with crabs preserved in formalin, which softens the carapace, thereby removing the original hard-soft criterion. Carapace width, sex, presence of eggs and carapace condition were recorded on size-frequency sheets aboard the vessels on which samples were taken.

On the Harengus trips, bottom samples for sediment type were taken with a 1/100 m² spring-loaded grab; bottom temperatures and salinity samples were obtained with a Nansen reversing bottle and thermometer. Positions for all stations were obtained by Decca, chain 9 in the west end of the Magdalen Shallows, chain 6 in the east end.

A sample of 135 crabs, taken on the Harengus cruise in August 1967 in the Gaspe area, was preserved in 10% formalin for later examination of stomach contents and gonad condition of male crabs. Stomach contents were examined in water under a binocular microscope (powers 10x to 42x); generally food organisms were identifiable only as belonging to a large taxonomic group, except for one species of amphipod which was identified. Maturity of male crabs was judged by the degree of development and thickness of the vas deferens. Presence

of eggs on female crabs was thought to be a sufficient criterion for maturity, as effectively 100% of female crabs in the upper fourth of the size range for females were egg-bearing.

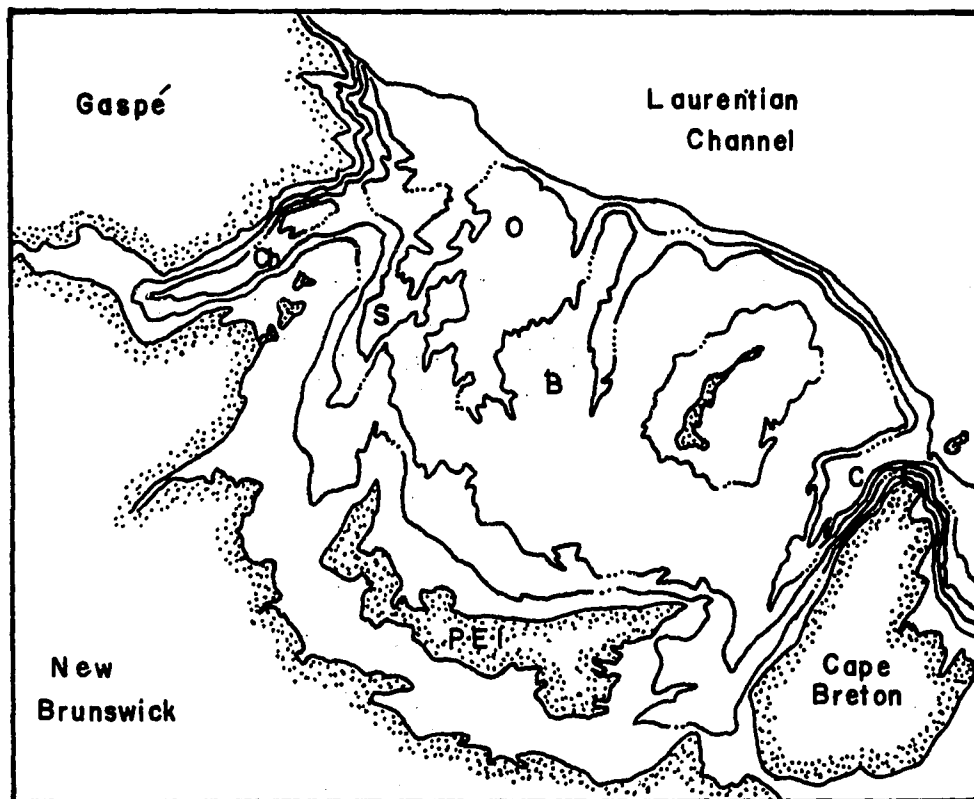
Two hundred and ten crabs were tagged in the Grande-Rivière area during fishing operations in August 1966. The tag used was a 14" length of bright yellow polyvinyl plastic "spaghetti" material, with a notice of reward for return, the address of the St. Andrew's F.R.B. station, and a tag number printed on. The tag was wrapped around the carapace between the cheliped and first walking leg and tied with a reef knot on the dorsal side of the carapace. The tag did not appear to interfere with the animal's movements. It was expected that tags of this sort would be of use only in showing short-term movements of the animals, as they would be effectively lost at the first moult following tagging.

An Edgerton underwater camera, mounted in a streamlined housing (designed and built at the F.R.B. St. Andrew's Biological Station) attached to the headrope of the Harengus otter trawl, was used to take bottom photographs on some stations of the August 1967 cruise. The camera takes photographs automatically every 12 seconds, and as mounted on the #35 otter trawl covers approximately 60 square feet of bottom area in each photograph.

Physical Features of the Magdalen Shallows

The southern part of the Gulf of St. Lawrence, bounded by lines running from the northeast tip of the Gaspé peninsula to the Magdalen Islands, and from the Magdalens to Cape North of Cape Breton Island, is called the Magdalen Shallows. The bathymetry of the Shallows is described in an excellent paper by Loring and Nota (1966); one of their figures is reproduced as figure 5. The bottom of the Shallows is generally flat, at an average depth of less than 40 fathoms. The flat bottom is interrupted by two prominent banks, Orphan Bank off the Gaspé coast and Bradelle Bank west of the south end of the Magdalen Islands, and by the Magdalens themselves. Minimum depth on the banks is 25-30 fathoms. The Shallows are further interrupted by a series of valleys, the most prominent of which occur at the western and eastern ends of the area. These are called by Loring and Nota the Chaleur Bay trough, which extends from Chaleur Bay out to the edge of the Shallows, where the slope into the Laurentian Channel begins, and the Cape Breton trough, running parallel to Cape Breton 10-15 miles offshore. Other valleys in the Shallows are the Shediac trough, roughly continuing the west end of Northumberland Strait and joining the Chaleur Bay trough west of Orphan Bank, and the Western and Eastern Bradelle trough, separated by Bradelle Bank.

Year-round water temperature distributions with depth for the the Gulf of St. Lawrence, and bottom temperatures on the Magdalen Shallows, are described by Lauzier (1957a and b), two of whose figures are given in illustration (figures 6 and 7). In general, in summer the water of the Gulf consists of three layers. A warm upper layer, sunwarmed, gives way at about 25 fathom depth to a layer of very cold water (-1° to 0°C), formed in situ during the winter. The deepest layer starts at



- Ch Chaleur Bay Trough
- S Shediac Trough
- O Orphan Bank
- B Bradelle Bank
- C Cape Breton Trough

Figure 5. Bathymetry of the Magdalen Shallows, after Loring & Nota (1966).

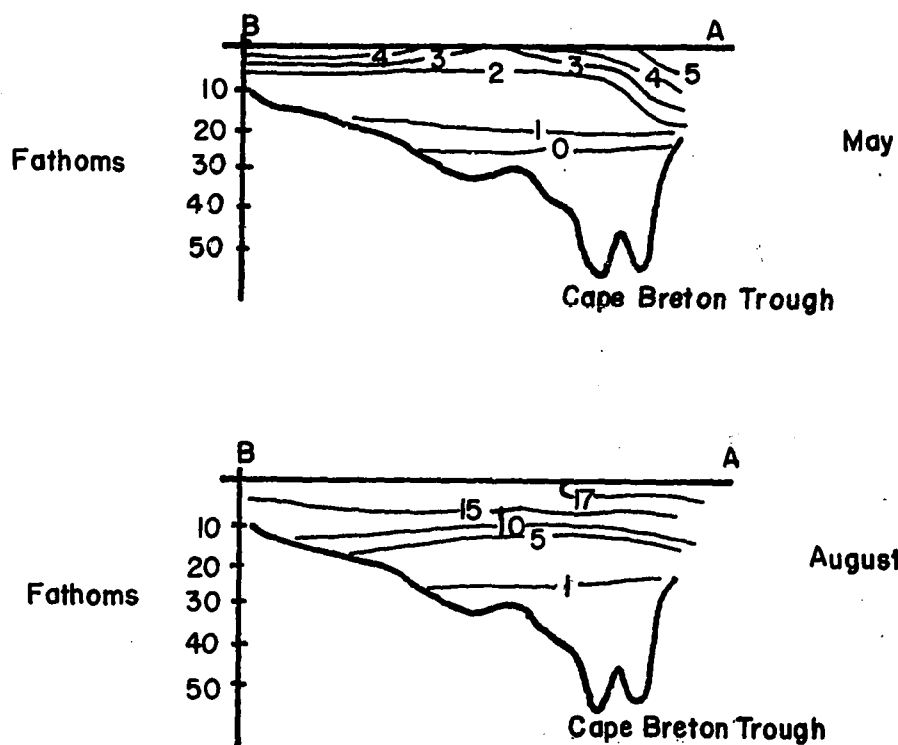


Figure 6. Spring and summer temperature distributions between Chéticamp and the Magdalen Islands. Transect location shown in Figure 2.

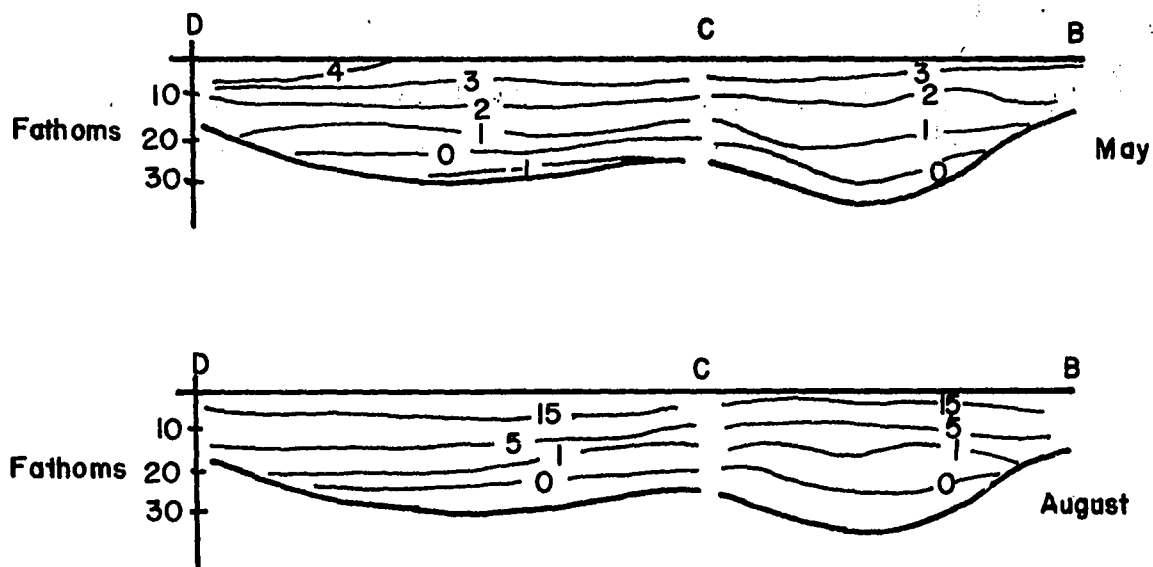


Figure 7. Spring and summer temperature distributions between North Point, P.E.I., Bradelle Bank, and the Magdalen Islands. Transect location shown in Figure 2. Both figures after Lauzier (1957).

about 80 fathoms depth, and has a temperature of about 5°C ; water of this deep layer flows in through Cabot Strait from the Atlantic. Effectively only the first two layers are found in the Magdalen Shallows, with the warm deep water being found only in the deeper parts of the troughs extending in from the Laurentian Channel. During fall, the top layer of the Gulf water cools to the same temperature as the intermediate water, allowing mixing before freezing occurs in January. The ice remains until April or May, and mixing is possible in early spring, after the ice breakup, before summer establishes stratification of the water again. In the deeper parts of the troughs, where some deep water influence is felt, temperatures of up to 3°C may occur, but in the main all of the Shallows bottom deeper than 25 fathoms is covered throughout the year by water of constant temperature lower than 2°C .

Distribution of sediments in the Shallows is described by Loring and Nota (1966), and a map of sediment cover produced by Loring (1964) is reproduced as figure 8. It will be noted from the map that muddy sediments are concentrated in the trough areas, while the shallower regions, i.e. most of the Shallows area, are covered principally with larger sands and gravels. Loring and Nota describe the depth-sediment correlation in some detail for the area between the Magdalen Islands and Cape Breton, indicating that pelite (fine sediment of diameter less than 0.05 mm.) is the dominant sediment at depths over 45 fathoms, while grain size increases with decreasing depth. Sediments on the slope from the Shallows are treated briefly by Loring and Nota; they state that gravel and rock predominate north and northwest of the Magdalens, which area would probably include some of the slope. In general, it might be expected that the Laurentian current would sweep small sediments off the slope except in areas protected from its direct effect.

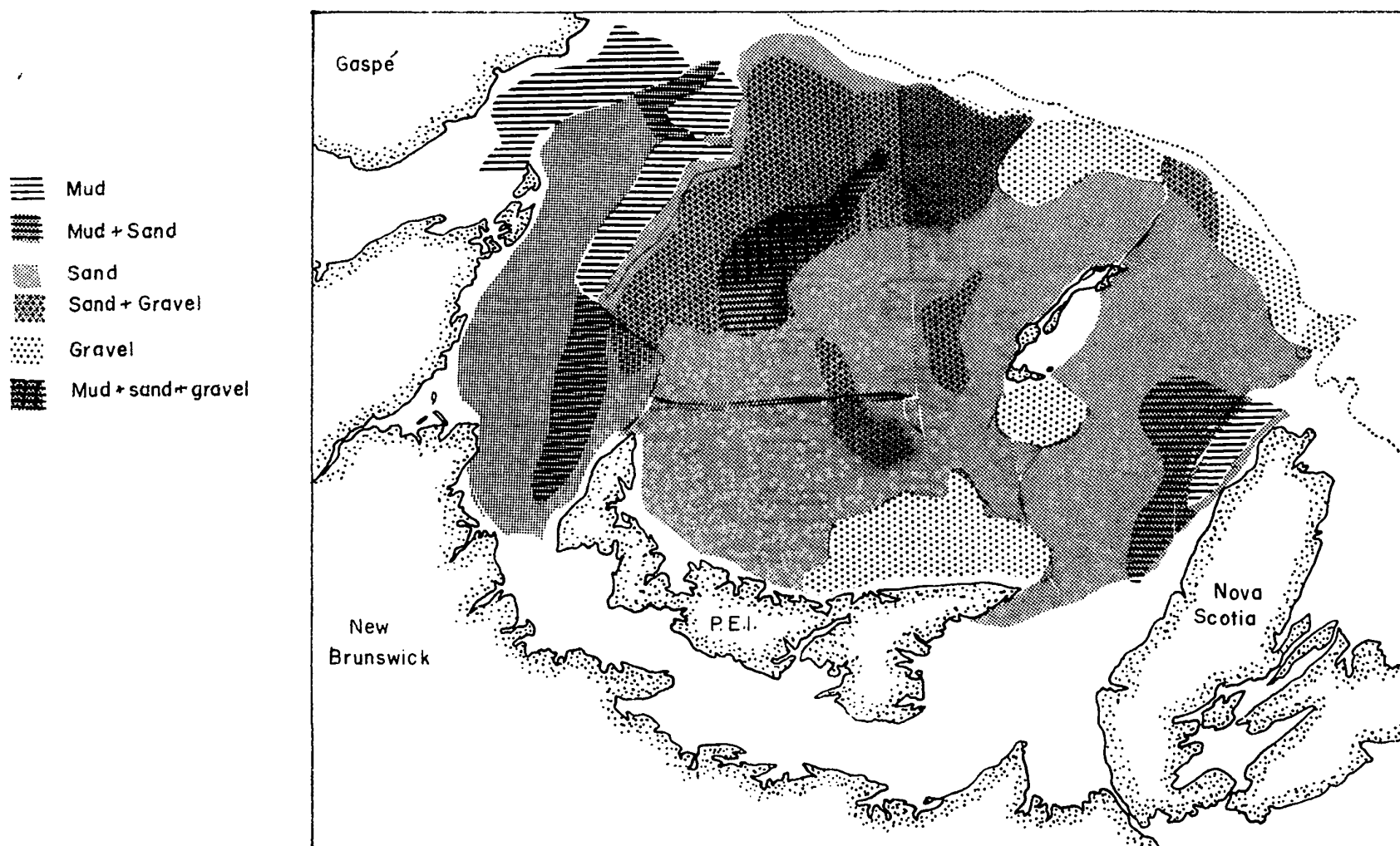


Figure 8. Bottom sediments on the Magdalen Shallows.

Distribution of Chionoecetes opilio

Distributional data are drawn from two principal sources, the Harengus cruise of August, 1967, and from the records of the Groundfish Investigation of the St. Andrew's Biological Station of the Fisheries Research Board of Canada. Because of the variety of conditions under which the latter samples were taken--different trawl types, time of day, etc.--the data were not expected to give any measure of comparative abundance in different areas, but to show only presence or absence of C. opilio. Numbers in any one sample were small, usually less than ten crabs. The Harengus samples were more uniform--gear type was constant and time of day noted--so were used to show comparative abundance in various area. Again, numbers taken per tow were small, but as lumped in the tables seem to show significant trends with various environmental factors.

Geographical distribution of C. opilio in the Magdalen Shallows is shown in figure 9. Raw data were taken from all sources, and were used to show only presence or absence of C. opilio in any sample. With the data as presented on the map, some indication of relative abundance may be had from the amount of shading in the square for each area, while the size of the square, indicating number of samples in the area, gives a measure of the reliability of the abundance estimate. Crab finds are concentrated in three chief areas on the map--the Chaleur Bay and Shediac troughs at the western end of the Shallows, and the Cape Breton trough at the eastern end. In both areas, C. opilio occur close to or outside of the 50-fathom contour; although it is found in shallower water, concentration is not so great. Some are also present in the slope area and in the deeper waters of the Laurentian Channel, particularly at the western end of the Shallows, but finds in these deeper waters are

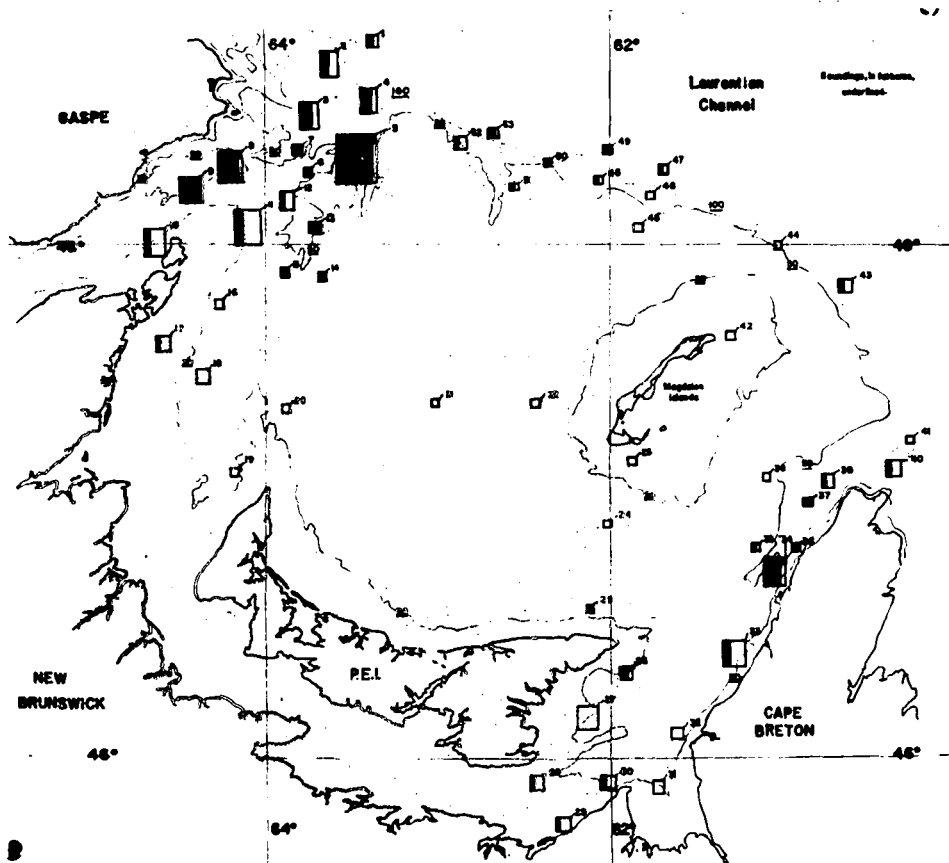


Figure 9. Distribution of *C. opilio* in the Magdalen Shallows. Size of square is proportional to number of tows in that area; amount of shading is proportional to number of tows yielding *C. opilio*. Numbers of tows and successful tows for each area are given in Table 3.

Point	# Tows	# With <u>C. opilio</u>	Point	# Tows	# With <u>C. opilio</u>
1	4	2	28	5	2
2	7	3	29	4	1
3	8	5	30	4	1
4	9	5	31	3	1
5	14	13	32	5	0
6	2	2	33	20	8
7	2	2	34	9	6
8	9	7	35	3	1
9	9	8	36	3	3
10	7	1	37	3	3
11	11	3	38	2	0
12	5	1	39	5	2
13	3	3	40	5	1
14	2	2	41	2	0
15	3	3	42	2	0
16	2	0	43	8	1
17	6	1	44	5	0
18	5	0	45	2	0
19	4	0	46	3	0
20	1	0	47	5	1
21	1	0	48	2	1
22	1	0	49	2	1
23	1	0	50	2	1
24	1	0	51	2	1
25	2	1	52	3	2
26	4	3	53	4	1
27	9	0			

Table 3. Number of tows and successful tows in areas shown in figure 9.

scattered and concentration is not so great as in the troughs. It is unfortunate that data were not more extensive for the central area of the Magdalen Shallows; however, the map suggests that C. opilio is at least not numerous in this area. Data supplied by the Fisheries Research Board were for cruises on which some stations had yielded C. opilio only, so that cruises on which none were taken were not available for this study; it is possible that cruises into the central area of the Shallows may have been taken, and if so would have yielded no C. opilio. No reports of catches of the crab in this central area have come to the writer, from fishermen or research personnel. Thus, although this species may occur in the central area of the Shallows, it is unlikely that it is present either in large concentrations, or extensively over the area.

Comparison of the distribution map with that of bottom sediments suggests a degree of correlation of presence of C. opilio with mud, or mixed sand and mud, bottoms. This is particularly noticeable in the Chaleur Bay-Shediac trough area, where a tongue of sand bottom separates the two troughs. Abundance of C. opilio in the tongue area is much less than that in either of the troughs, whose bottoms are predominantly mud. The bottom in the central area, where C. opilio appears to be scarce, is largely sand, with gravel mixed in, with occasional patches of mud. Bottom types in the slope and Laurentian Channel areas are not mapped, but as noted earlier, large sediments or exposed bedrock probably predominate on the bottom in the slope area.

The presence and numbers of C. opilio with varying depth are given in tables 4 and 6. Catch figures in the tables for the Harengus work are for the maximum daytime catch on individual stations on which the otter trawl was not damaged. Damage was frequent and apparently

Depth (fathoms)	Number of stations	Number with <u>C. opilio</u>	% with <u>C. opilio</u>
0- 9	4	-	-
10-19	38	10	26
20-29	50	10	20
30-39	35	3	9
40-49	18	5	28
50-59	35	17	49
60-69	26	14	54
70-79	10	-	-
80-89	19	4	21
90-99	6	1	17
100-109	6	-	-
110-119	1	-	-
120-129	7	2	29
130+	9	-	-

n = 264

n = 66

Mean depth, all stations, 50.40 fms.

SD = $\pm$ 32.41 fms.Mean depth, stations with C. opilio, 49.39 fms.SD = $\pm$ 24.93 fms.

t = .0982, p > .9

Table 4. Presence of C. opilio with varying depth,
from F.R.B. data.

Temperature °C	Number of stations	Stations with <u>C. opilio</u>	% with <u>C. opilio</u>
<-1	2	-	-
-1 to -0.1	7	3	43
0 to 0.9	63	18	29
1 to 1.9	42	15	36
2 to 2.9	28	5	18
3 to 3.9	17	2	12
4 to 4.9	22	-	-
5 to 5.9	12	1	8
6 to 6.9	-	-	-
7 to 7.9	6	-	-
8 to 8.9	4	-	-
9 to 9.9	8	1	13
10+	11	1	9
	n = 222	n = 46	

Mean temperature, all stations, 2.97°C

SD = $\pm 3.02^{\circ}\text{C}$

Mean temperature, stations with C. opilio
= 1.64°C

SD = $\pm 2.15^{\circ}\text{C}$

t = 12.269, p < .001

Table 5. Presence of C. opilio with varying bottom temperature, from F.R.B. data.

Depth (fathoms)	Number of stations	Mean catch per station	Range in catches
20-29	1	0	-
30-39	-	-	-
40-49	8	10.0	1-23
50-59	11	16.1	0-51
60-69	4	16.3	10-21
70-79	-	-	-
80+	1	80	-

Table 6. Catch with depth, Harengus 1967.

Temperature, °C	Number of stations	Mean catch per station	Range in catches
< 0	2	10	6, 14
0 to 0.4	17	12.0	0 - 51
0.5 to 1.0	2	7	1, 13
1.0 to 2.0	2	40	0, 80

Table 7. Catch with bottom temperature, Harengus 1967.

affected trawl efficiency considerably; night catches are consistently 5-6 times higher than day catches at the same station (and would have been used in the tables if night sets had been made at all stations) so were excluded. From table 4, the t-test between means of the depth distributions for all stations and for stations at which C. opilio were found shows no significant difference; further, no real trend in percent of stations with C. opilio at varying depths is apparent, except for a clumping around 50-69 fathoms depth. It is apparent from table 6 that depths were not widely sampled on the Harengus cruise, with 19 of 25 stations in depths of 40-59 fathoms; thus no significant trend can be shown in variation of abundance with depth, except for the one haul at 80 fathoms, producing 80 crabs.

Variation in catch of C. opilio with temperature is summarised in tables 5 and 7. Table 5 shows a significant difference between the means of the temperature distributions for all stations and stations with C. opilio, with the crab occurring at a lower mean temperature than that of the sampled areas. Further, the percentage of stations at which C. opilio were found decreases with increasing temperature. As with depth, a wide range of temperatures was not sampled on the Harengus cruise, with 17 of 23 stations for which temperatures are available being in the 0° to 0.4°C range. No conclusion can be drawn from table 7 except that C. opilio are found at temperatures of -0.2 to 2.0°C , with no indication that higher temperatures are inimical to the animal's life. The apparent very high mean catch at 1° - 2°C was wholly contributed by the catch of 80 animals at 80 fathoms, discussed below.

Squires (1966) found a depth range of 50 m. to 300m., with 80% of all finds occurring between 100 and 225 m., and a temperature range of -1.1°C to 2.3°C , 80% of finds coming between -0.5 and 0.5°C .

Thus, depth and temperature distributions for C. opilio are much the same inside and outside the Magdalen Shallows, for Canadian east coastal areas.

Although depths and bottom temperatures were not sampled widely on the Harengus cruise, different bottom sediments were fairly well distributed among the stations sampled, with 15 stations on predominantly muddy bottoms and 8 on sandy bottoms. Table 8 summarises catch of C. opilio per one-hour trawl set in relation to bottom type. It will be evident that numbers of C. opilio are greater on muddy bottoms than on sandy, and when a t-test is applied to the difference between the means of catches on sand and mud the difference is found to be significant at the .05 level. It will be further noted from the table that sand bottoms on which there is some admixture of mud show a somewhat higher number of C. opilio than the unmixed sand bottoms. Table 9, summarising finds of C. opilio with varying bottom types from F.R.B. data, confirms this conclusion--16 of the 17 stations on which C. opilio were taken for which bottom type was recorded were on mud bottoms. Bottom type on the fishing grounds off Cheticamp, Cape Breton, was sampled on the Harengus September 1967 cruise and found to be predominantly mud. Brunel (1960) notes that the species is found on mud bottoms, and suggests that the crabs burrow into the mud as an escape from predators, particularly at the times of moulting and mating, when they are vulnerable to predation. Some evidence to support this hypothesis was found in this study, which will be discussed later.

	Number of tows	Total <u>C. opilio</u>	Mean catch per hour
Mud	with gravel	1	16
	unmixed	11	198
	with sand	3	101
	TOTAL ON MUD	15	315
Sand	with mud	4	34
	with mud and gravel	1	13
	unmixed	3	14
	TOTAL ON SAND	8	61
			21.0 SD=±20.8
			8.5
			13
			4.7
			7.6 SD=±9.5

$$t = 2.1134 \quad d.f = 22$$

$$.05 > p > .01$$

Table 8. Abundance of C. opilio with bottom type, Harengus 1967.

Bottom type	Number of stations	Number with <u>C. opilio</u>	% with <u>C. opilio</u>
Mud	55	16	29
Sand	16	-	-
All others	15	1	7

Table 9. Presence of C. opilio with bottom type, F.R.B. data.

Changes in distribution with time

Two hundred and ten male C. opilio were tagged and released on the fishing grounds off Grande-Rivière (hatched area in figure 2) in August 1966. Although fishing effort for C. opilio was concentrated in this area, adjacent grounds are fished with otter trawls and gill-nets, both of which take crabs, so that crabs which had moved some distance would stand a chance of being picked up. In fact, 19 of the 20 tag returns were made by crab fishermen, maximum movement from the release point for these 19 animals being 4 miles over $11\frac{1}{2}$ months. All 7 animals at liberty for more than 9 months, for which accurate return positions are known, moved significantly, from 1 to 4 miles. Six of these movements were southerly, while the seventh was to the north. The recapture point for the single trawl-caught crab was given as "Miscou Bank, 30 fathoms"; this would require a movement of about 10 miles SSE, over a period of 13 months. Several writers have suggested that C. opilio populations in the Bering Sea move to shallow water in spring for hatching of eggs and for copulation. Returns from the tagging program were not possible during months of ice cover (late December to May in 1966-67), so the possibility of a similar phenomenon here cannot be discounted. The concentration of returns around the release point, however, suggests that migration on a large scale between release and recapture does not occur unless it is so highly organised that animals return to a small home area after migration.

Diurnal variations in numbers caught with trawls and gill nets were noted by Brunel (1960, 1961, 1962), night catches being 1.5 to 3 times as great as those during the day. Similar fluctuations were noted on the Harengus cruise, where night catches averaged 5 times as great as day catches on the same station, and in the trawl fishery in 1966; these

are summarised in tables 10 and 11, and in figure 10. Brunel postulated that the crabs burrowed into the substrate and remained inactive during daylight hours, becoming active at twilight and feeding during the night; during daylight, the trawl would pass over the buried animals. Bottom photographs taken in 1967 support this hypothesis. Figures 11 and 12 show crabs photographed on a day and night set. The former animal appears to be partly buried, with its leg outlines indistinct and carapace region pressed to the ground, while the latter is up on leg tips, with a clear shadow indicating separation of the carapace from the bottom. This is the only picture available from night sets, so the behavior may not be typical; no crabs photographed during day sets cast clear shadows, and three of the seven shown are partly buried. Brunel's hypothesis appears to be correct, and indeed provides the only reasonable explanation for the higher night-time catches. Accordingly, it is assumed that the trawl catches measure the general activity of crab populations sampled, when day and night catches are compared. The ultimate purpose of this behavior is probably predator avoidance--cod and other groundfish are known to eat C. opilio (F.R.B., unpublished data), and cannibalism is indicated from examination of stomach contents (see below). In discussing reduced catches in May, Brunel suggests that burying occurs during both day and night at that time, when moulting activity is apparently greatest, since the newly-moulted crab is particularly vulnerable to predation. Its proximate cause appears to be simply light level. Figure 10 suggests some lag in the control, i.e. maximum activity is not reached until several hours after sunset, and minimum activity not re-established until after sunrise; and this would be expected of a living system under direct control of some cycling factor.

Date	Time →												
	1800	1900	2000	2100	2200	2300	2400	0100	0200	0300	0400	0500	0600
June 22-23						160	160	160	35	35	35		
June 29-30			400	400	425	425	425	460	460	1040	1040	1040	
July 2-3						350	350						
July 5-6				140	140	140	530	530	500	500	500		
July 12-13				510	510	520	520	670	670	1020	1020	1020	
July 18-19					540	540	540	540	540	540			
July 20-21			350	350	860	860	860	980	980	980			
July 25-26				400	400	500	500	500	525	525			
July 27-28					325	325	325	530	530	230	230	230	
Total/hour			750	1700	3550	3820	3860	4320	4240	4870	2825	2270	
N.			2	5	8	9	8	8	8	8	5	3	
Avg./hour			325	340	444	536	482	552	530	609	565	763	
			↑ Sunset									↑ Sunrise	

Table 10. Trawl catchability of *C. opilio* with time of night. Date from fishery off Grande-Rivière, 1966.

		Time →																		Trawl catching rates (crabs/ hour)		
		0700		0900		1100		1300		1500		1700		1900		2100		2300			2400	
	19	21	21	1	1	16	0	0	21	21	10	10	-	-	258	258	103	100				
	38	19	12	23	23	13	9	9	8	6	7				224	224	100					
		38	1	10	16	5	5	51	51	7	25				23	23	40					
				1	10	2	2			25					61	61						
				12	1											40						
																103						
Total/hour	57	78	34	47	51	36	16	60	80	59	42	10			566	709	243	100				
N.	2	3	3	5	5	4	4	3	3	4	3	1			4	6	3	1				
Avg./hour	29	39	11	7	10	9	4	20	27	15	14	10			142	118	81	100				

Trawl
catching
rates
(crabs/
hour)

Table 11. Trawl catchability of C. opilio with time of day. Date from Harengus cruise, August 1967

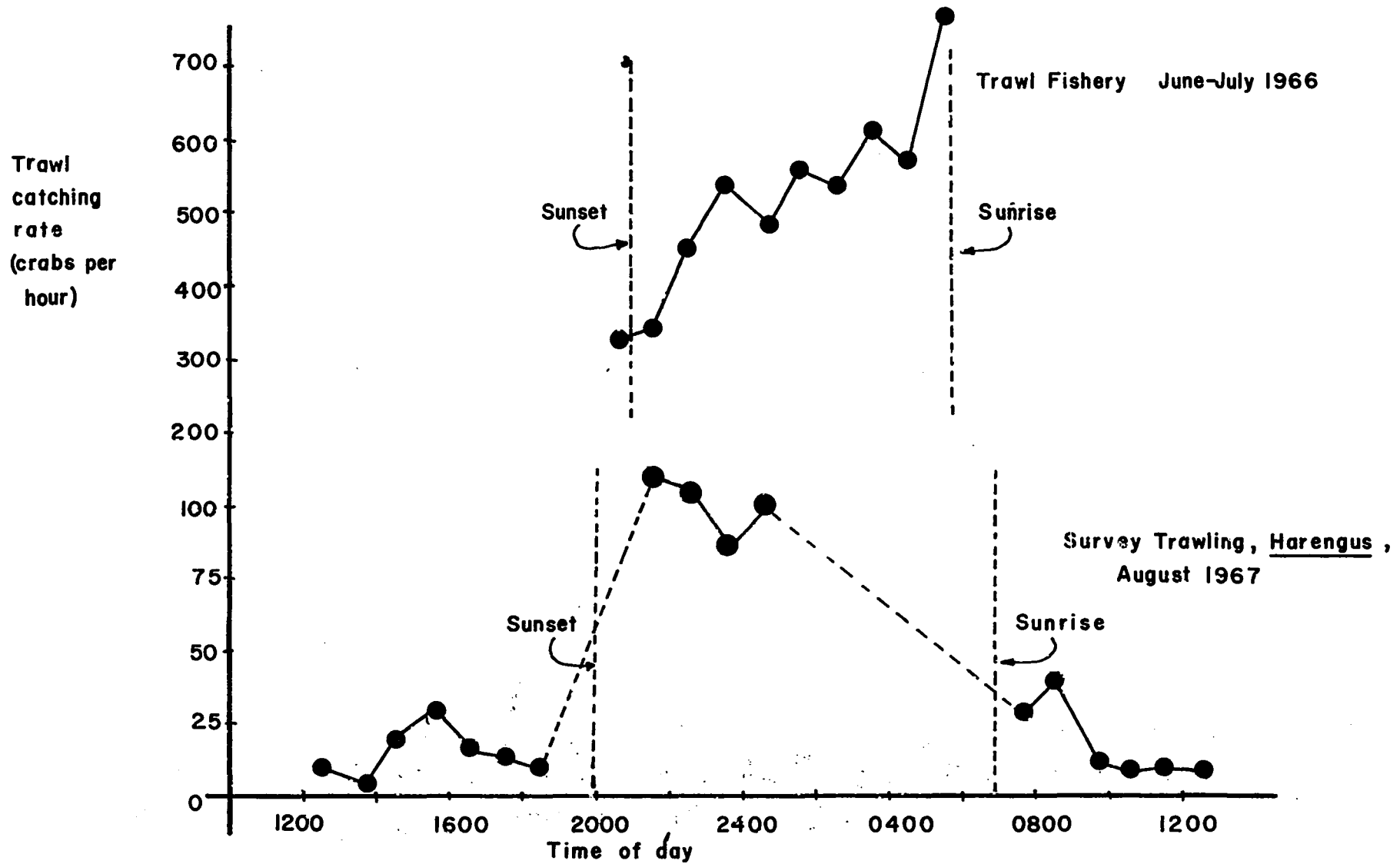


Figure 10. Trawl catchability with time of day.



Figure II. Crab photographed by underwater camera during daytime
trawl set.



Figure 11. Crab photographed by underwater camera during daytime trawl set.

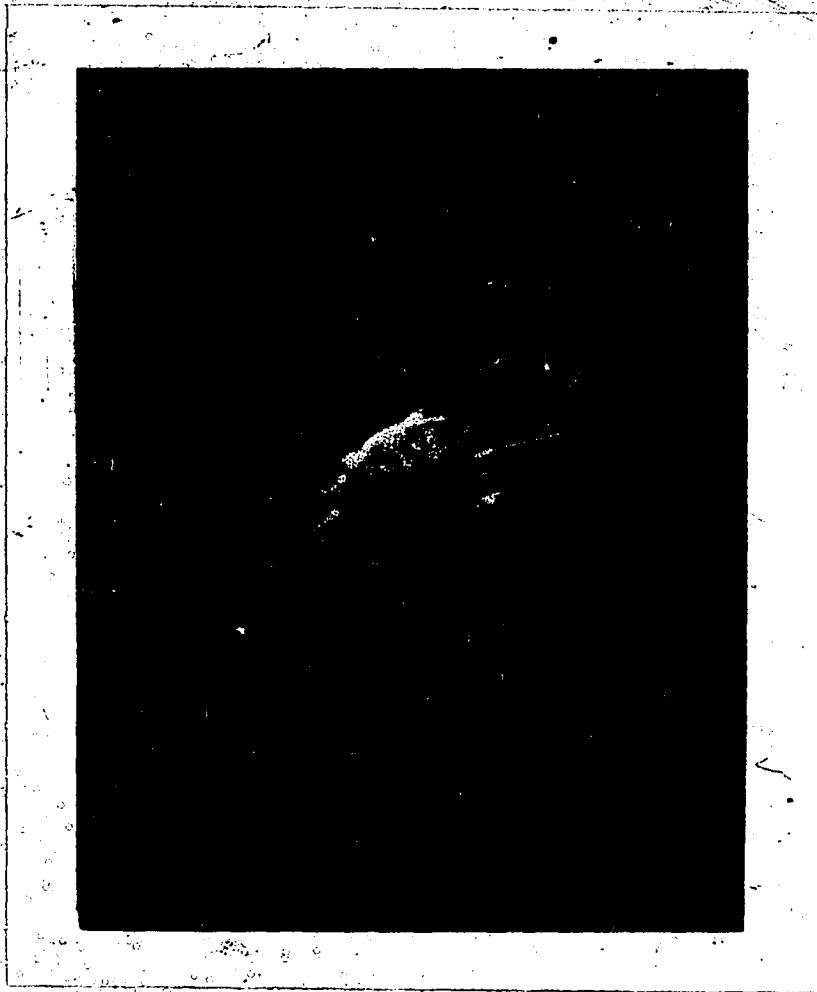


Figure 12. Crab photographed during night trawl set.



Figure 12. Crab photographed during night trawl set.

If this diurnal behavior is simply controlled by light, one would expect the difference between day and night catches to decrease as depth was increased, other conditions being equal. Table 6 shows the largest daytime catch (80 animals) occurring at the greatest depth sampled (80 fathoms was the average depth of the two, maximum being 92 fm.), and a slight tendency towards higher catches at greater depths. These findings tend to support the light hypothesis for control of diurnal activity differences.

Local distribution

It will be noted from tables 6 and 7 that catches of C. opilio under the same conditions of temperature and depth, in approximately the same area (many of these were inside the 50 fm. contour off Grande-Riviere) were quite variable, ranging for example from 0 to 0.4⁰ C. This suggests that crabs aggregate to some extent on a smaller scale than the wide geographical areas over which these conditions occur. Figure 13 shows numbers of crabs caught against depth for a number of 24-hour trap sets off Cheticamp, in the Cape Breton trough. Depth ranged from 60 to 84 fm., over which little range in temperature or bottom type would be expected, except perhaps for a slight warming in the deeper water. The average depth of the trough here is 50-60 fm., the greater depths of the figure occurring in a narrow (less than $\frac{1}{4}$ miles) gully within the trough. The observations could not be repeated, but an observer on the fishing vessel reported some months later that catches were better in the gulley and that the fishermen were setting traps there in preference to the shallower trough area. This suggests that a small deep area is somehow beneficial to the life of C. opilio; perhaps predation is less likely because of the greater darkness during the day, or because of an accumulation of soft sediments.

Since the traps were set one day and hauled the next, decreasing light level at greater depth does not explain the increased catch. If numbers of crabs were constant over this 60-84 fathom range, night would provide the darkness necessary for those at shallower levels to come out of the mud, move about, and enter the traps, and in this case catches at 60 and 84 fathoms would be expected to be closer to equal. An understanding of the catching dynamics of the crab traps is necessary to this argument; although this subject is not within the domain of this

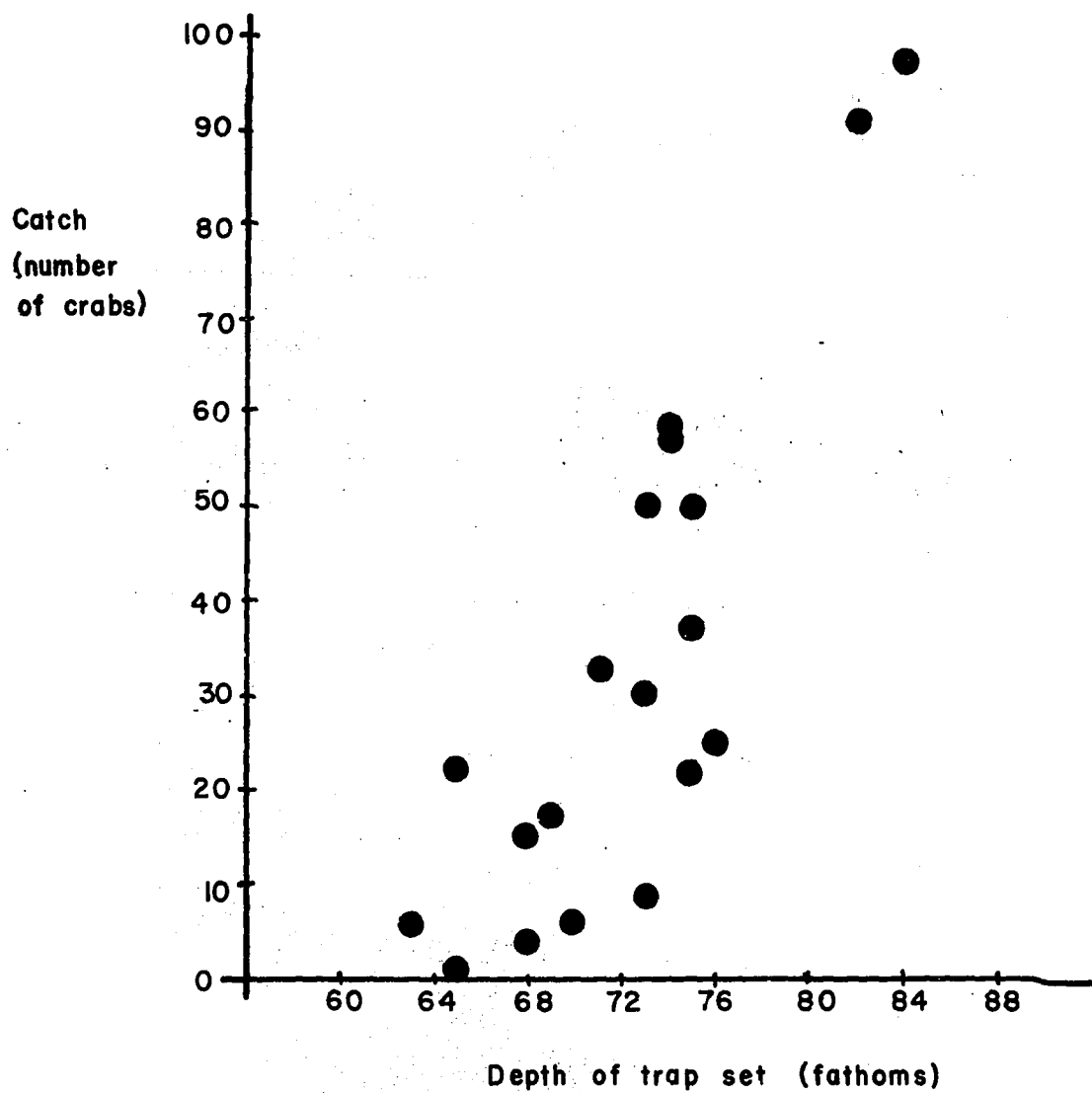


Figure 13. Local depth distribution of C. opilio within Cape Breton Trough.

work, it should be noted that observations in the Chaleur Bay Trough indicated that a trap takes as many commercial (i.e. large hard-shelled) crabs as it is going to take when set one day and hauled the next. In sets of longer duration, catches become slightly smaller, perhaps because of a lessened attractiveness of the bait and a slow rate of escape over time. One must conclude, therefore, that the difference in trap catches with depth shown here reflects an actual difference in numbers present, and not a difference in light levels causing an apparent difference in numbers.

Density of *C. opilio* populations

An estimate of *C. opilio* density can be made using the trawl-swept area and catch. The swept area is calculated from the effective width of the trawl--80 feet for the #35 otter trawl (P.J.G. Carrothers, personal communication)--and the length of the set, found from Decca positions. The trawl catch represents only part of the whole population over which the trawl has passed, as will be discussed later, so a density estimate arrived at in this way will be low. The factor by which the estimate should be increased to give greater accuracy is not known. Night catches are used for the estimate, as these are consistently 5-6 times greater than day catches in the same area, and represent actual densities better than day catches. Various estimates are summarised in table 12, from catches on the Harengus cruise and from catch data of the experimental trawl fishery for *C. opilio* in June and July 1966. The latter figures are for commercially useful crabs only, i.e. hard-shelled animals of carapace width greater than 100 mm., and thus include less of the population than the Harengus figures, which include all animals caught. The swept area for the fishery samples is less than that for the Harengus figures, as the boat used was somewhat smaller and covered less distance in an hour's trawling. The fishery figures give an average catch for 23 tows in the area bounded by the 50-fathom contour off Grande-Rivière, and the maximum catch (the average of the three greatest catches) in the same area. Two stations in this area, and one inside the 50-fathom contour at the southern end of the Shediac trough, are quoted for the Harengus cruise.

<u>Sample description</u>	<u>Swept area</u>	<u>Catch</u>	<u>Square feet per crab</u>
1. Trawl fishery off Grande-Rivière (area shown in figure 2).			
A. average catch	(80 x 2.5 x 5300) 1.06 x 10 ⁶ sq. ft.	530	2010
B. maximum (avg. of three)	1.06 x 10 ⁶ sq. ft.	990	1070
2. <u>Harengus</u> <u>August 1967</u>			
A. Chaleur Bay Trough	(80 x 3 x 5300) 1.272 x 10 ⁶ sq. ft.	224	5680
B. Chaleur Bay Trough	1.272 x 10 ⁶ sq. ft.	103	12400
c. Shediac Trough	1.272 x 10 ⁶ sq. ft.	258	4930
3. Edgerton camera Shediac Trough	(200 x 62) 12,400 sq. ft.	4	3100

Table 12. Estimates of density of C. opilio populations.

Density estimates from the fishery are somewhat higher than those from the Harengus, and the difference in population terms would be greater, since the Harengus samples include a greater proportion of the population than do those from the fishery. The difference may be due to a decrease in numbers caused by the crab fishery, which was intensive over a small area during the ice-free months from July 1966 to August 1967. The fishery figures on which estimates are based may, however, be unreliable, and are certainly more so than the Harengus sample figures, so that comparisons between the two are not likely to be valid. The Harengus data are reliable, and give a good estimate of the numbers of animals large enough to be taken into the trawl.

Density estimates from bottom photographs would be very good, since the camera "samples" every animal on the bottom, except that the total area covered in photographs on one tow is quite small. One photograph covers about 62 square feet; for a one-hour tow with a photograph every 12 seconds, possible total area photographed is 18,600 square feet, as compared to 10^6 square feet with the trawl. Four C. opilio were photographed on a daytime set in the Shediac trough area for which density was estimated; number of exposures was 200 instead of the 300 possible (due to some malfunction). This gives a density of 1 crab for 3100 square feet, roughly comparable to the 1 per 4930 square feet estimated from trawl sampling. Many such sets would have to be made to yield figures large enough to be useful.

Size Frequencies

Size frequency histograms for two trap and two trawl samples of C. opilio are given as figures 14-17. Figure 14 is for crabs subsampled from several crab catches with an unmodified #35 otter trawl; figure 15 treats the whole catch of C. opilio for the August, 1967 Harengus cruise, using the modified #35 otter trawl; while figures 16 and 17 are for one day's catch using a small-mesh and large-mesh 6' x 6' trap respectively. It is evident from figures 15-17 that neither gear is adequate for sampling over the whole size range--below 100-110 mm. carapace width the number caught declines sharply, and effectively no crabs of either sex are taken below 60 mm. The first histogram, figure 14, does not show the decrease in sampling efficiency below 100 mm.-- this is probably because the sample measured was a subsample of several trawl catches, while the other three were for an entire catch. The bias against small sizes is, in the case of the trawl, probably due chiefly to the footrope's passing over small animals; few animals of less than 60 mm. width were caught even in the modified otter trawl with its $\frac{1}{2}$ " mesh liner, which would certainly retain even very small animals which got into the trawl. In the case of the traps, selection bias against small animals is probably due to their secretiveness in the face of numbers of larger crabs; feeding studies indicate that C. opilio is cannibalistic to some extent, and attack of small animals by larger ones is likely. The difference in size range for the samples from the two types of trap is very probably a result of the different mesh sizes used, with the trap used in the Grande-Rivière area having the smaller mesh, and yielding greater numbers of small animals. A shortage of soft-shelled animals is evident in the trap samples as compared to trawl samples; again, timidity of the soft-shelled animals is the probable cause, as these are

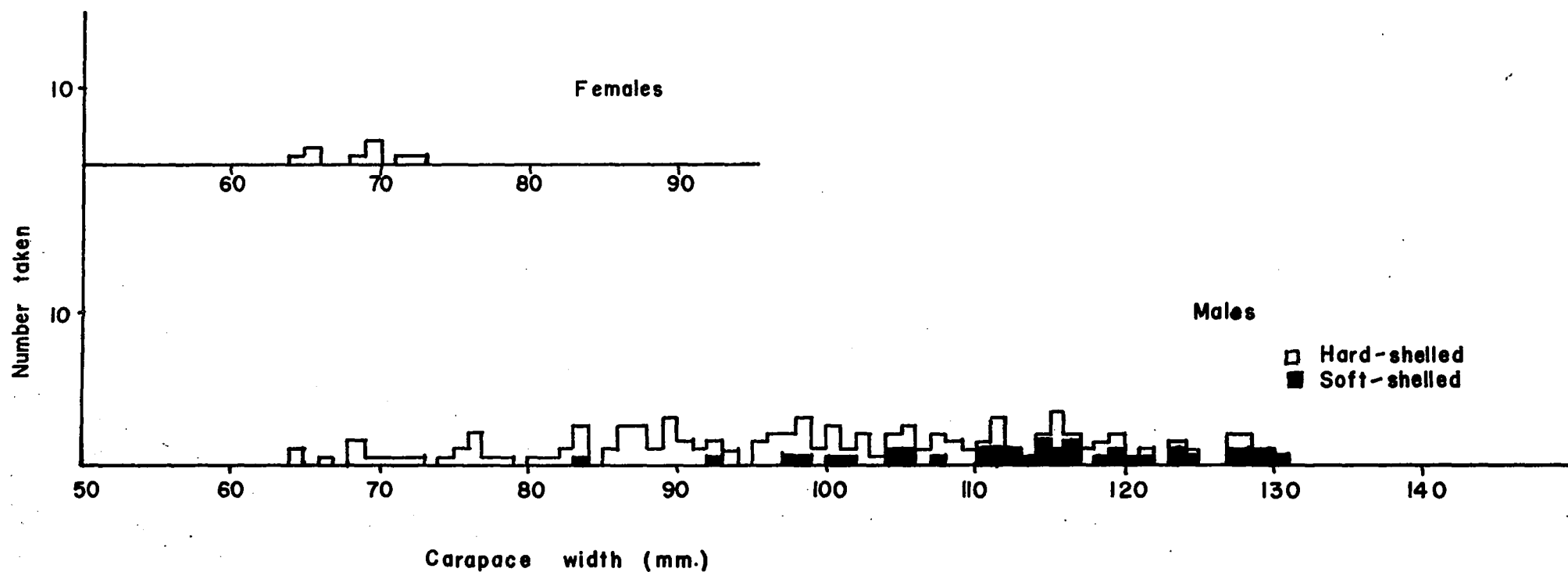


Figure 14. Size frequencies of C. opilio trawl-caught in fishing area off Grande-Rivière (shown in Figure 2) in July 1966.

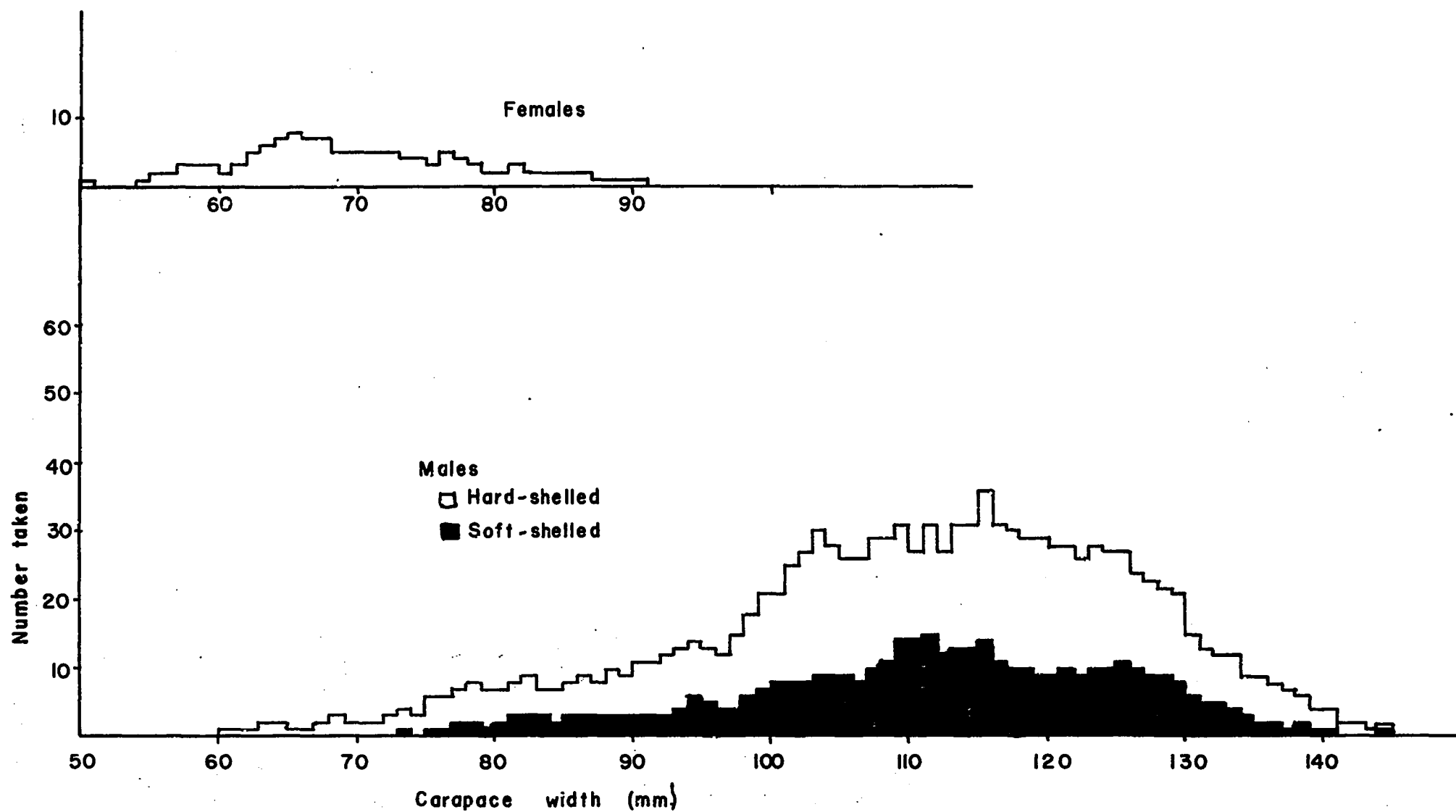


Figure 15. Size frequencies of *C. opilio* trawl-caught in the Chaleur Bay and Shediac Troughs during August 1967.

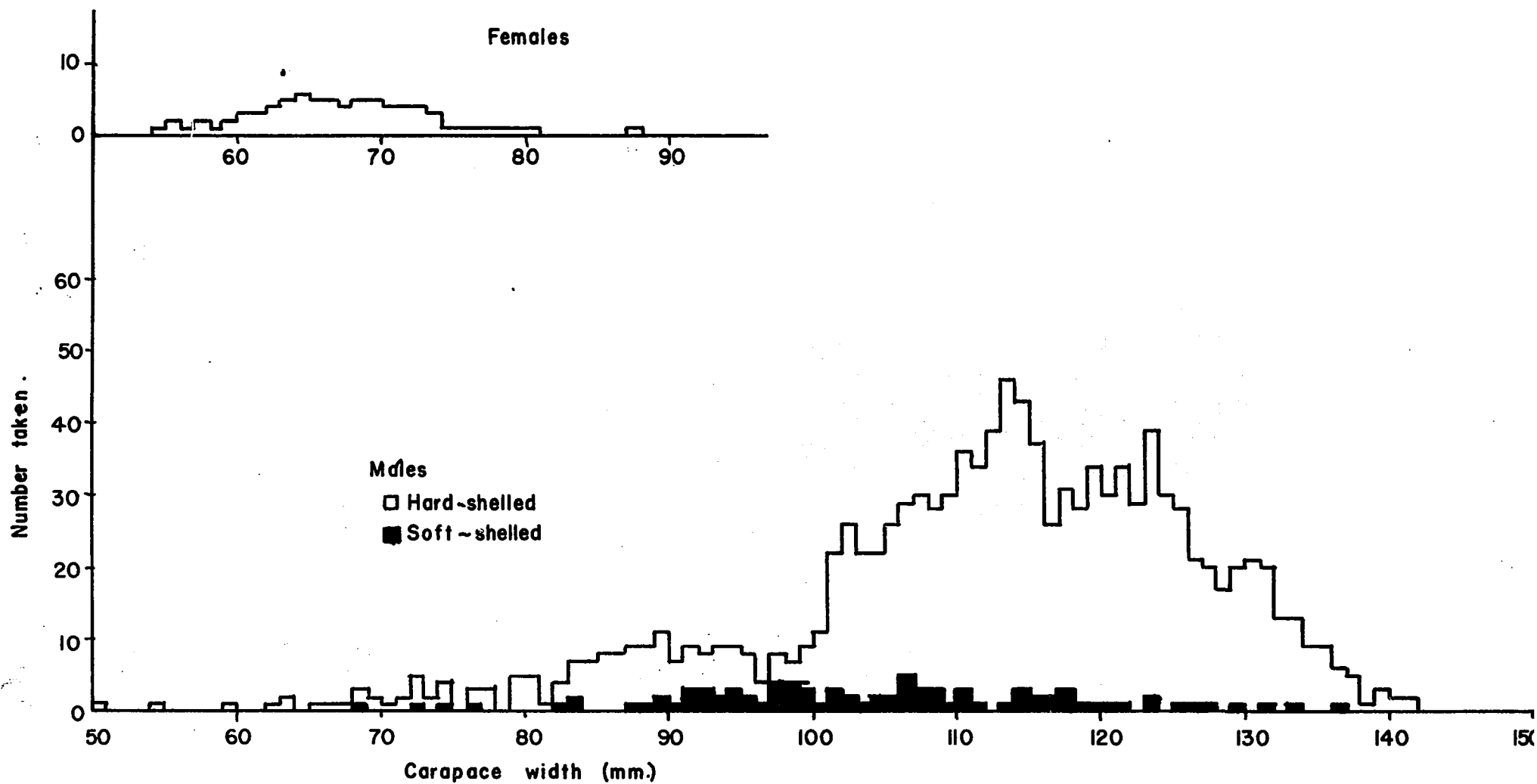


Figure 16. Size frequencies for C. opilio trap-caught in fishing area off Grande-Rivière, August 19, 1966.

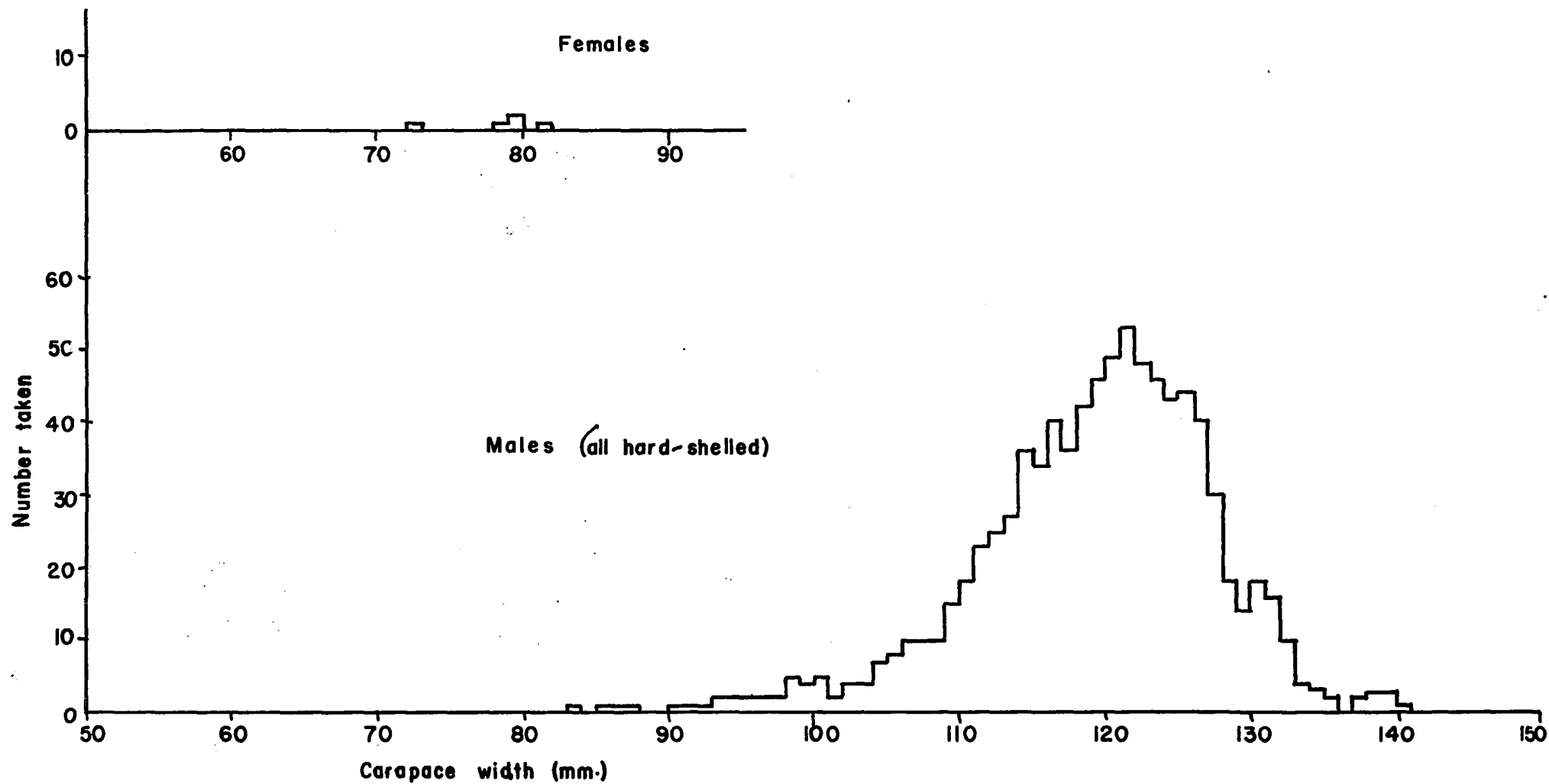


Figure 17. Size frequencies for *C. opilio* trap-caught in fishing area off Chéticamp (shown in Figure 2) July 1967.

very vulnerable to attack by hard-shelled crabs. There are indications that activity affects trawl catches, as well as those of the trap; Brunel (1960) suggests that soft-shelled animals bury themselves in the bottom mud in order to escape from predators, and the trawl would be more likely to pass over such buried crabs than over exposed animals. Thus, the trawl may have a bias against soft-shelled animals. This factor might further affect catches of small animals, who may also bury themselves in the mud to escape predation.

Squires (1965) obtained an average carapace length for males of 25 mm. (equivalent to 30 mm. carapace width) and a maximum length of 61 mm. (73 mm. width), as compared to our average width of about 115 mm., and maximum of 149 mm. Likewise, his average length for females was 23 mm. (28 mm. width) and maximum 45 mm. (54 mm.). However, Squires' samples were selected at sea for measurement on shore, and he explains (p. 110) that in order to fit more animals into collecting containers at sea, small crabs were preferred to larger ones.

Females are, of course, affected by the bias of both sampling methods against small animals, but the presence in the samples of more females than males of the same size is interesting. The governing factor may well be different activity of males and females of these sizes; the females sampled, being in the top section of the female size range, may have more in common behavioristically with large males than with males of their own size, with similar requirements for food and thus for movement. If this is so, females would be more inclined to enter traps in search of food, and to be active on the bottom and thus subject to capture by the otter trawl.

An alternative explanation for the lack of small sizes in the samples is that they simply are not present on the grounds sampled. The above discussion assumes a usual equilibrium population age structure, with animals of all sizes present, and the writer feels that this assumption is not unreasonable, since a population of C. opilio has existed for some years in the Chaleur Bay Trough off Grande-Rivière (at least since 1960, when it was first studied by Brunel). Further, it is probable that gear selection works to some extent in the manner described. It is nonetheless conceivable that, due to a large successful settlement of larvae some years ago, large sizes may have been unusually plentiful in 1966 and 1967 when this work was done. Further, bad years for settlement might have almost completely removed small C. opilio from this area. Some fishermen in the area (whose theories should not be given much weight) describe a great increase in numbers of the crab over the past four or five years, and our trawl catches in 1966 and 1967 were far greater than Brunel's in 1960, 1961, or 1962. Both Brunel (1960-62) and Squires (1965) took smaller animals in numbers at least more in keeping with a conventional population structure than did our gear, which was quite similar to theirs. The possibility exists, then, that the samples taken represent the true population fairly accurately, and that small C. opilio are at present very scarce, which would imply a population whose year-class numbers fluctuate considerably.

Growth of male crabs in the final moult

Two of the size frequency distributions for male crabs only, those of figures 15 and 16 (called, respectively, samples A and B in the following discussion) were plotted on probability paper and analysed following the method of Harding (1949). Preliminary analysis on the two whole samples, males and females, showed a simple bimodal distribution, one mode made up of females and smaller males, the other of the larger males. From this and the results of the following analysis it was concluded that male and female moulting in these size ranges were different and could be treated separately. The female size-frequencies shown here fall into simple unimodal normal distributions, not permitting further analysis. Sizes of male crabs were plotted in five mm. groups; the resulting probability plots are shown in figures 18 (sample A) and 19 (sample B). Each sample was assumed, in turn, to be a unimodal normal distribution (whose probability plot is a straight line), a bimodal normal distribution (whose probability plot takes the form of two straight sections connected by a point of inflection), and a trimodal normal distribution (three straight segments connected by two points of inflection). Harding's method enables one to separate the component normal distributions from a set of data distributed polymodally and to graph them; the method requires estimation of the position of the point(s) of inflection connecting straight sections of the original plot, and from this estimate the distribution is resolved graphically into its component parts. In figures 18 and 19, the x's represent the raw data; the estimated point of inflection, assuming a bimodal distribution, is marked; the straight lines above and below the original points represent the two theoretical normal distributions estimated from the raw data; and the curved line represents the theoretical bimodal distribution made up of the two theoretical normal distributions. This theoretical combined distribution

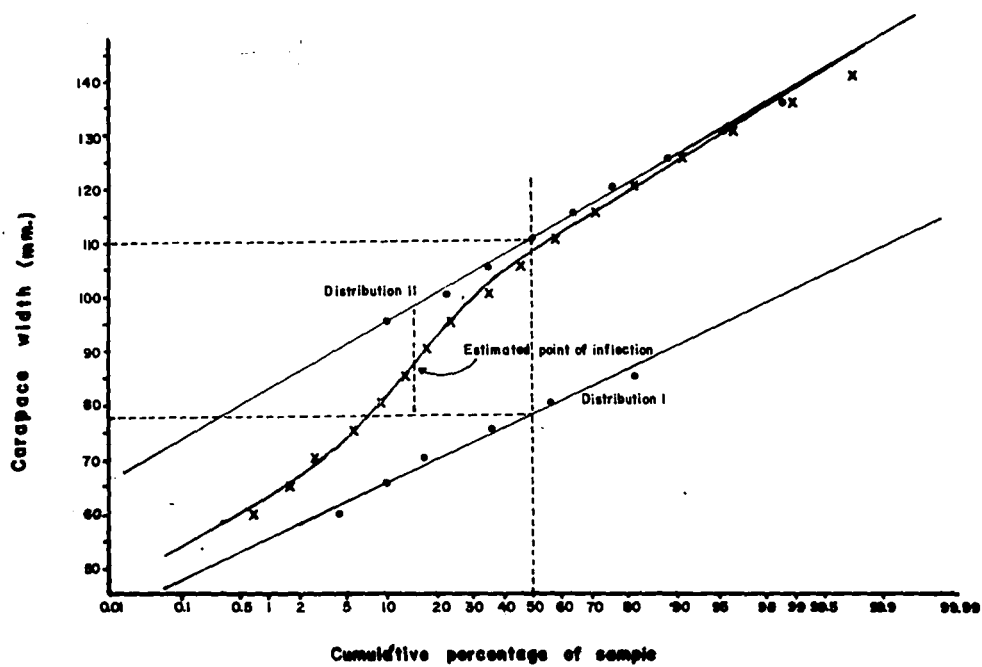


Figure 18. Sample A— from trawl-caught sample, Chaleur Bay Trough,
August, 1967.

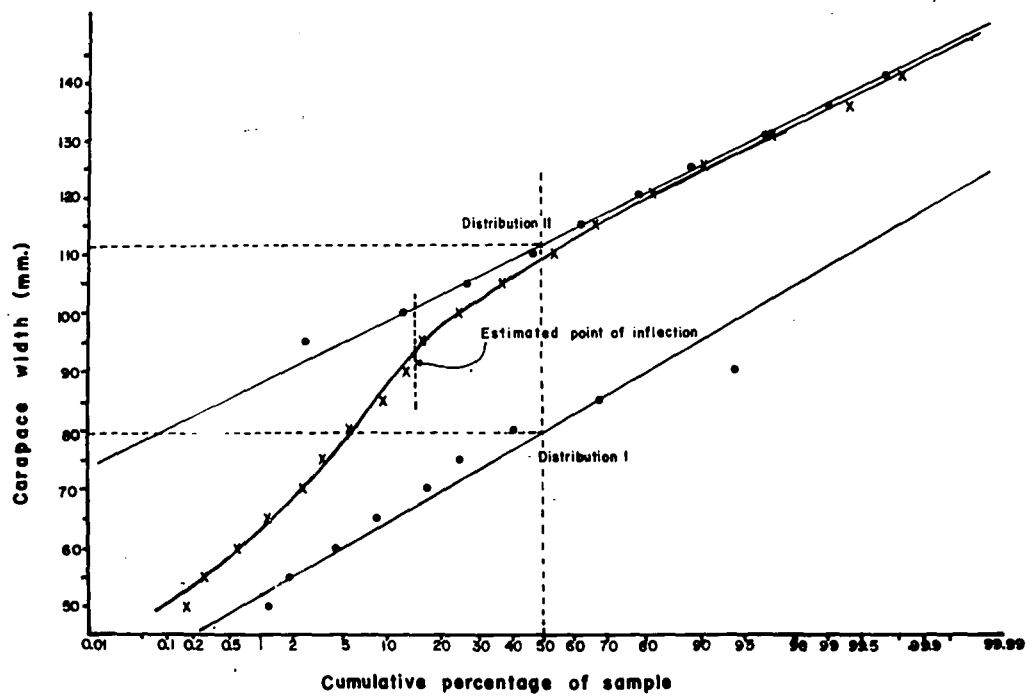


Figure 19. Sample B—from trap-caught sample, Chaleur Bay Trough,
August 1966.

Distribution A - from figures 15 and 18

<u>Model</u>	<u>Chi²</u>	<u>d.f.</u>	<u>P</u>
Unimodal	>100	17	<<.001
Bimodal	24.47	17	.10 > p > .05
Trimodal	24.23	17	.10 > p > .05

Distribution B - from figures 16 and 19

<u>Model</u>	<u>Chi²</u>	<u>d.f.</u>	<u>P</u>
Unimodal	>100	16	<<.001
Bimodal	41.25	16	< .001
Trimodal	80	16	<<.001

Table 13. Goodness of fit of various models to raw size-frequency data.

is then tested against the raw data by χ^2 , results of the various χ^2 tests performed in this analysis being given in table 13. Data from sample A fit to a bimodal model fairly well, with p between 0.05 and 0.10; the trimodal model fits insignificantly better. χ^2 values for all three models as applied to sample B are large, with p less than 0.001 in each case; the smallest occurred when the distribution was fitted to a bimodal model. The total χ^2 value was much affected by several high values of $\frac{(f_c - f_o)^2}{f_c}$, while the rest were small, of the order of 1. This reflected unevenness in the original data, and it was concluded that although these data could not be fitted well with any of the models, the bimodal model gave a working approximation to them.

The results of these analyses, as read from the probability plots, are summarised in table 14. For each sample, "distribution I" refers to the earlier moult-class, and "distribution II" to the later. The results from the two samples are in good agreement, suggesting that fitting of distribution B to the bimodal model was justified, and that the samples represent the situation in the population as a whole. Further, since the two samples were taken in succeeding years, constancy of population structure is suggested over this period.

The significance of these results depends, of course, on what sort of population is being sampled and how it is being sampled. A theoretical equilibrium crustacean population, under constant conditions, with eggs hatching at discrete intervals, has the size-frequency form of figure 20, with each normal curve representing a moult-class. Numbers per class decrease, mean size increases, and standard deviation increases, as one advances to higher classes, and this structure is the same at the same time in any year. If all sizes are present in the population sampled during this study, sampling effectiveness has had the form of

Sample	Distribution	% of sample represented	Mean carapace width	SD
A Trawl 1967	I	15	78 mm.	$\pm$ 11mm.
	II	85	110 mm.	$\pm$ 12mm.
B Trap 1966	I	14	80 mm.	$\pm$ 13mm.
	II	86	112 mm.	$\pm$ 11mm.

Table 14. Results from probability paper analysis of male size-frequency distributions.

figure 21. The result of superimposing the two curves, i.e. sampling the population, is figure 22, the sample. This is the form of the samples analysed above. Such a sample structure might also result from the population structure (derived from differential success of year-classes) and sampling effectiveness shown in figures 20a and 21a, or from some combination intermediate between the two population structures and sampling efficiencies shown. If our results are representative of the first situation, (figures 20 and 21), the estimated mean of the first distribution, and probably of the second, is too high, since smaller members of the distributions are not sampled as effectively as larger ones. This effect would be more pronounced on distribution I than on distribution II, so that the growth per moult estimate (distance between the means of the two distributions) would be too small. In the second situation (figures 20a and 21a), the estimate would be valid.

Remembering that these factors will affect this conclusion, it is concluded that male C. opilio in the final moult-class have a mean carapace width of about 111 mm., with a standard deviation of ± 11.5 mm., while those in the next-to-last class have a mean width of about 79 mm., and a standard deviation of about ± 12 mm. The growth per moult estimate is 31 mm., about 40%, of width in the final moult.

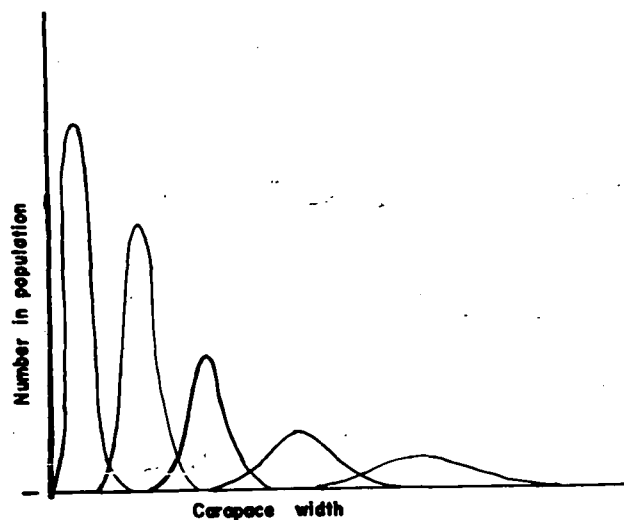


Figure 20. Size structure of theoretical crustacean population in which success of larval settlement is constant, and larvae settle at discrete intervals.

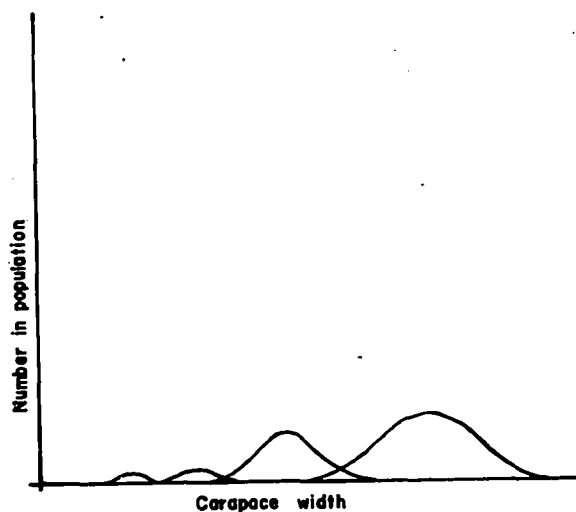


Figure 20a. Size structure of theoretical crustacean population in which settlement success varies; settlement occurs at discrete intervals.

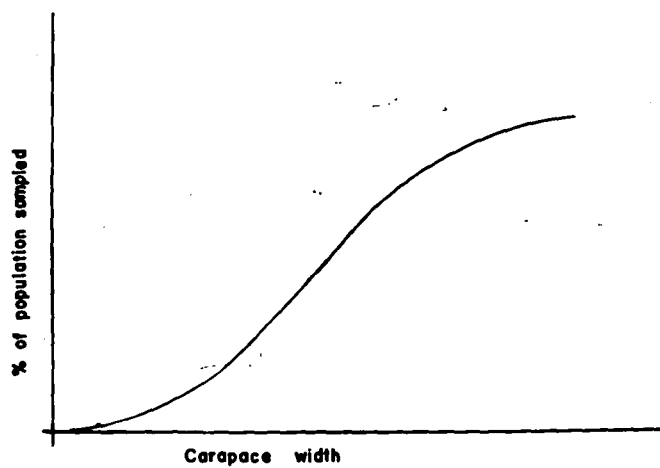


Figure 21. Curve of sampling efficiency when efficiency decreases with size.

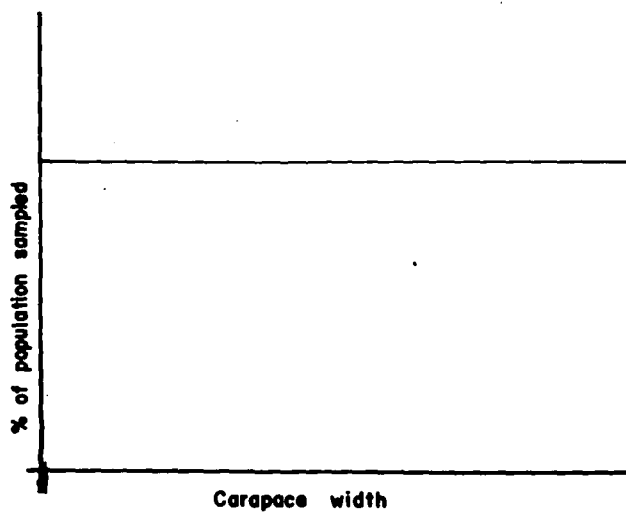


Figure 21a. Curve of sampling efficiency when it is independent of size.

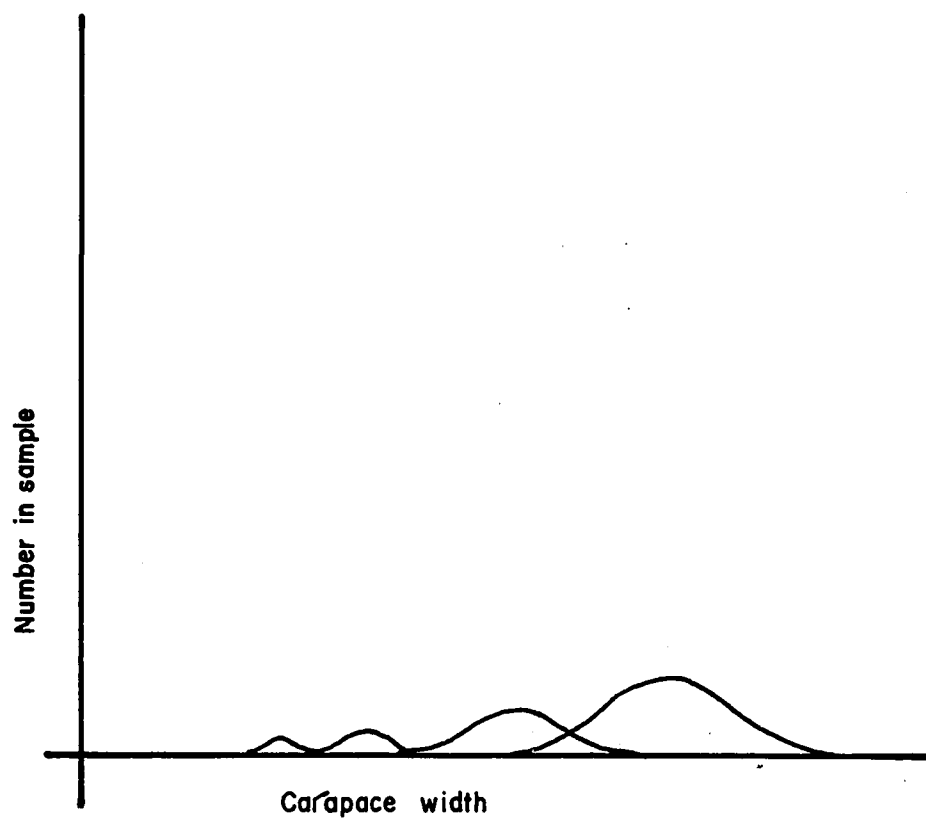


Figure 22. Size structure of theoretical sample found by combining Figures 20 and 21 or 20a and 21a.

Relationship of weight to size

One hundred and twelve male C. opilio of carapace widths between 63 and 147 mm. were weighed after being drained with shaking, anterior down, until water had stopped dripping from the carapace. Ninety-one of these, of carapace widths 104-147 mm., were weighed to the nearest gram; the other twenty-one, of carapace widths 63 to 102 mm. were weighed to the nearest half-ounce and the weights were converted to grams. All were at approximately the same stage of the moult cycle, having old carapaces with some growth of epibiotics. Weights are plotted against carapace widths in figure 23; the growth line in the figure was fitted by eye. From this figure and the preceding estimate of mean increment in carapace width in the final moult, one can estimate the mean weight increment in this moult as 350 gm. (600 gm.-250 gm.) or about 140%.

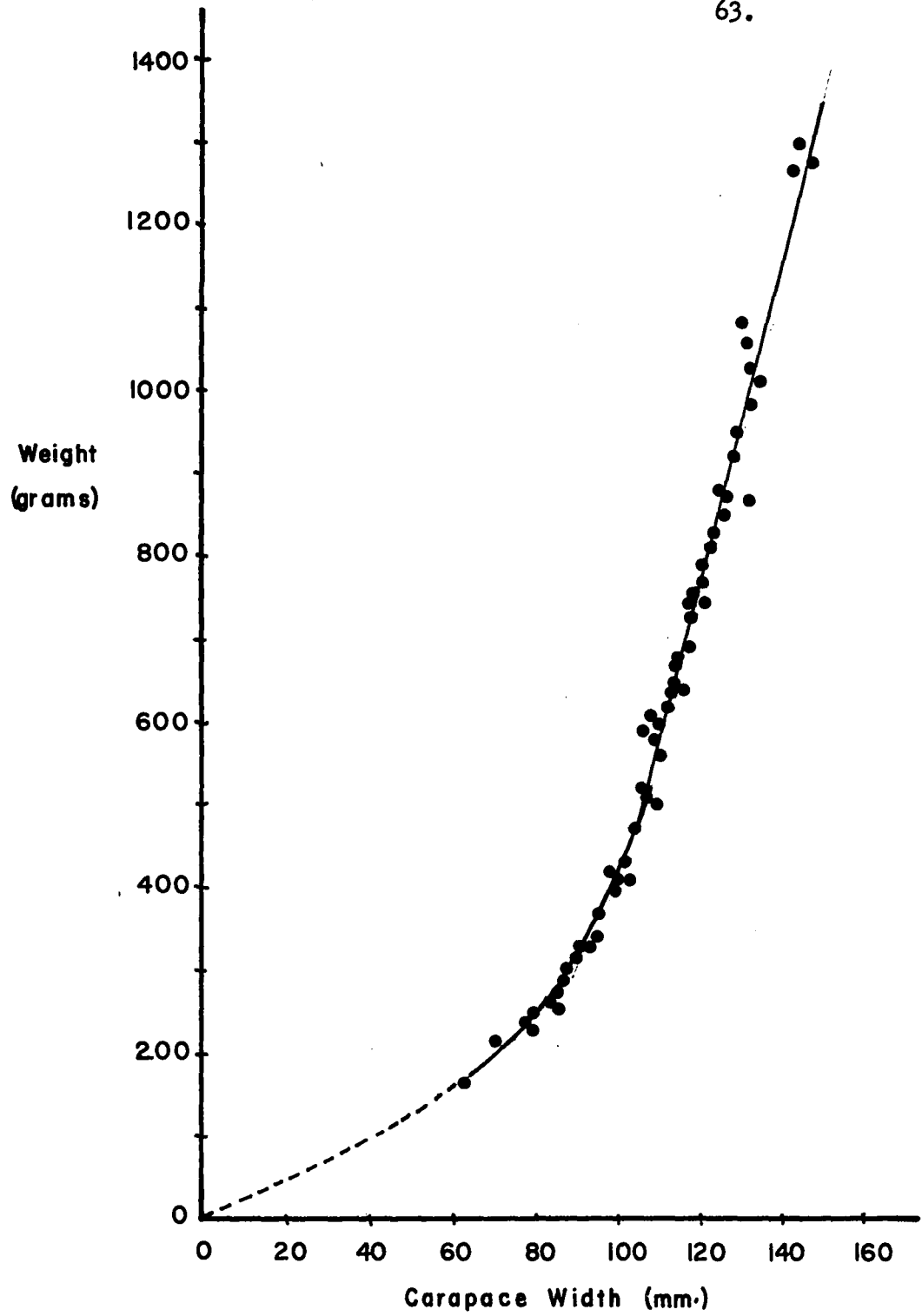


Figure 28. Size-weight relationship for male *C. opilio*.

Allometry

One hundred and thirty-eight males and 29 females from the Chaleur Bay Trough, and 181 males and 12 females from the Cape Breton Trough, were measured for allometry studies. The characteristics measured, besides carapace width which was used as index in the logarithmic plots, were claw length and width and width of the third abdominal segment (see figure 24). These were selected because of their possible relation to sexual maturity--large claws are frequently a secondary sex characteristic in male brachyura, while the abdomen is widened in mature female brachyura. Claw length and width were multiplied to give claw area. Claw area and abdomen width were plotted against carapace width on double-logarithmic paper for the two sexes in each area (figures 25-30), and the least-squares regression line was fitted to each set of logarithmic values (table 15, figures 31 and 32).

These sets of lines demonstrate three things. First, sexual dimorphism is shown clearly--at equal carapace widths, claw area of males is larger than that of females, while abdomen width is greater in the females. These characters are, then, sexual ones. Presence of a moult is shown in the graph of claw area vs. carapace width for males from the Chaleur Bay Trough, this sample being separated into two distinct groups, one extending from carapace width 80 mm. to the top of the size range, the other for animals of less than 112 mm. The correspondence of this separation to that shown by the size frequency analysis is clear; a rough mean for the larger size group can be had by adding the extreme size values (80 mm. and 144 mm.) and dividing by 2, giving 112 mm., corresponding to the 111 mm. found by Harding's method. The 50% overlap

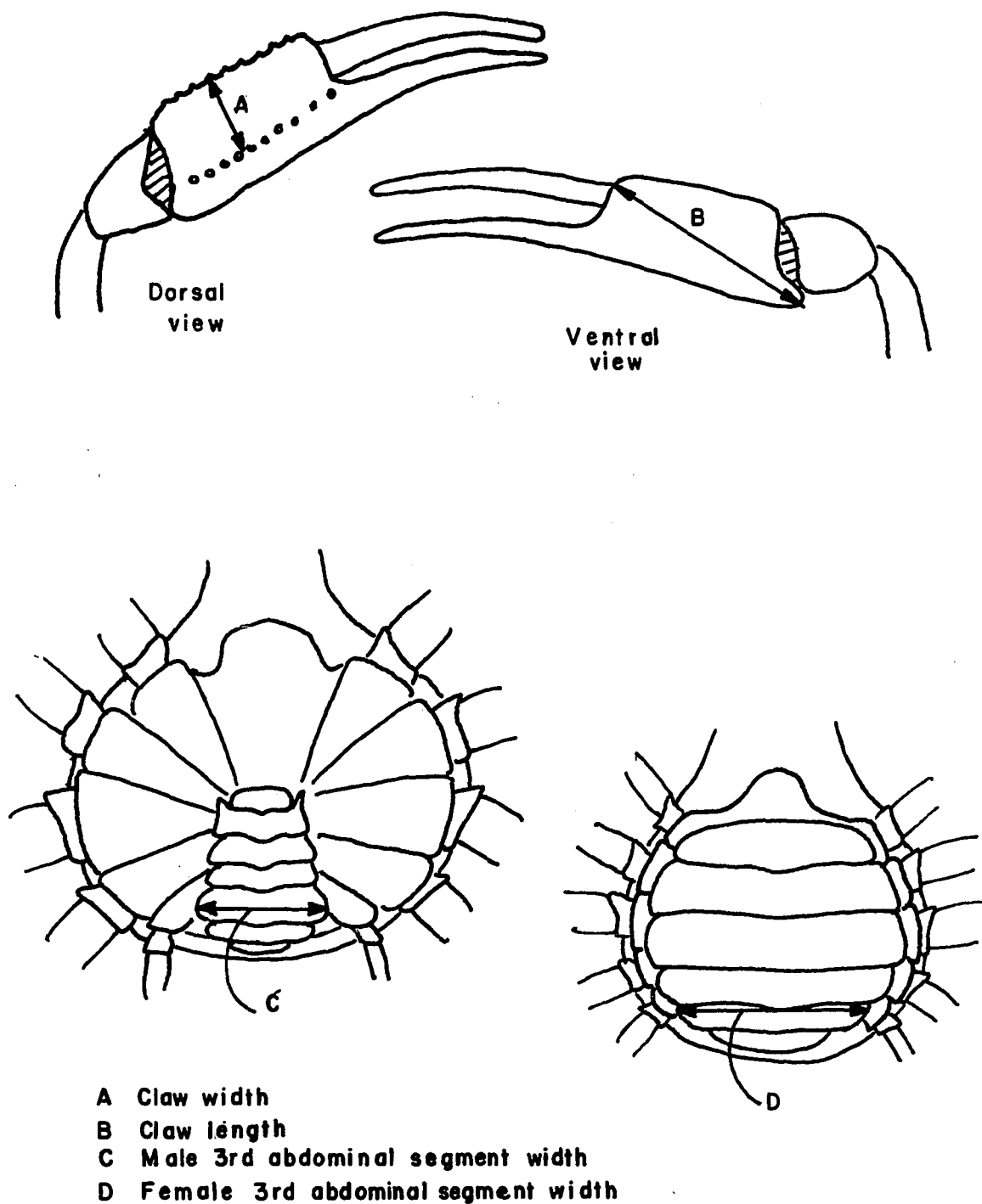


Figure 24. Characters measured for allometry studies.

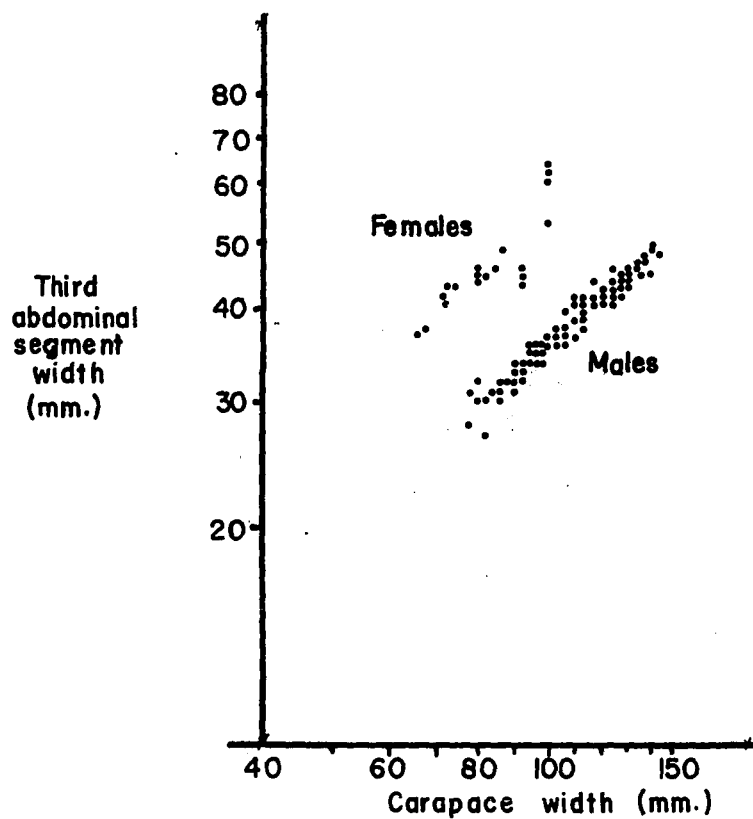


Figure 25. Allometric relationship of width of 3rd abdominal segment to carapace width—Cape Breton Trough.

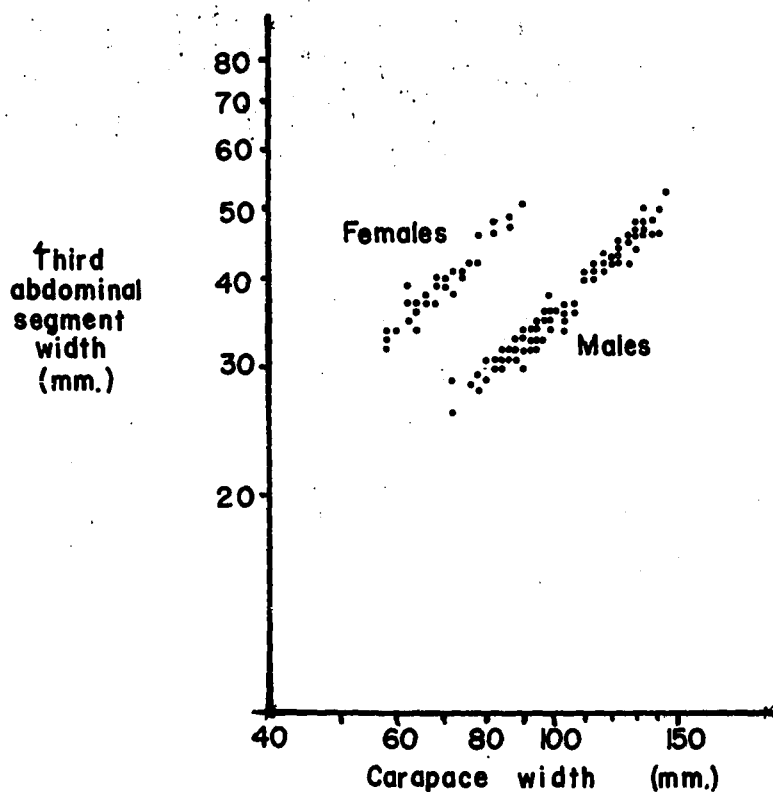


Figure 26. Allometric relationship of width of 3rd abdominal segment to carapace width—Chaleur Bay Trough.

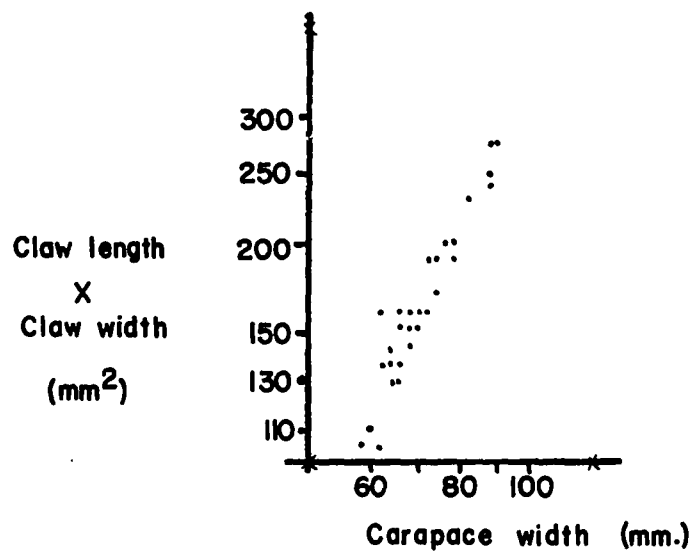


Figure 27. Allometric relationship of claw area to carapace width— females, Chaleur Bay Trough.

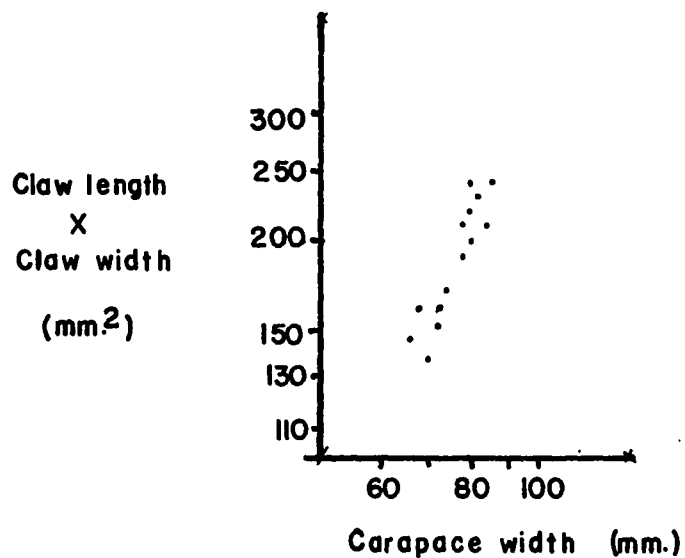


Figure 28. Allometric relationship of claw area to carapace width— females, Cape Breton Trough.

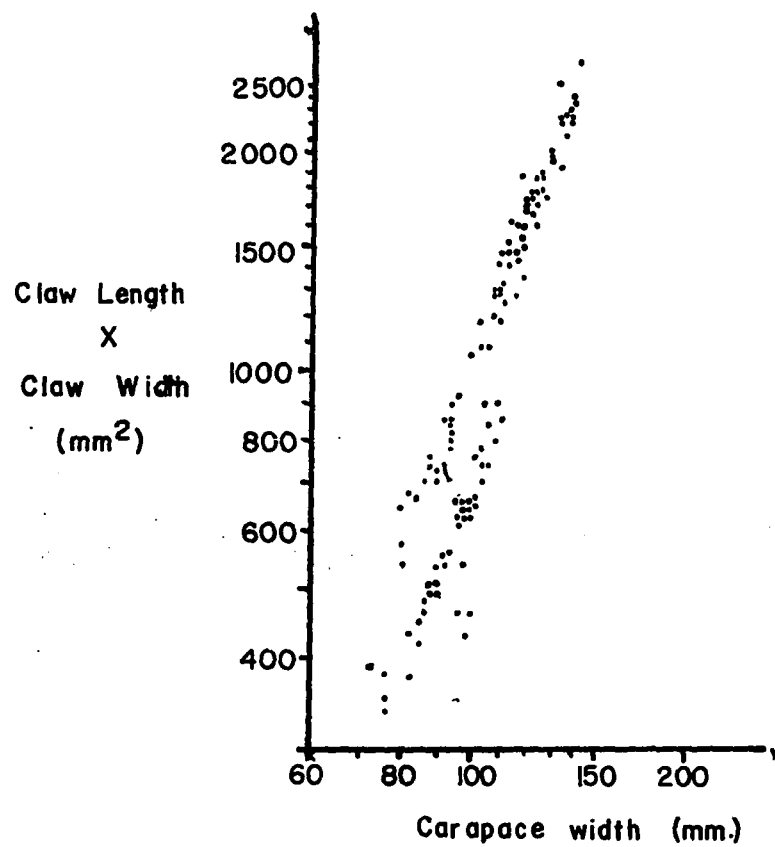


Figure 29. Allometric relationship of claw area to carapace width—males, Chaleur Bay Trough.

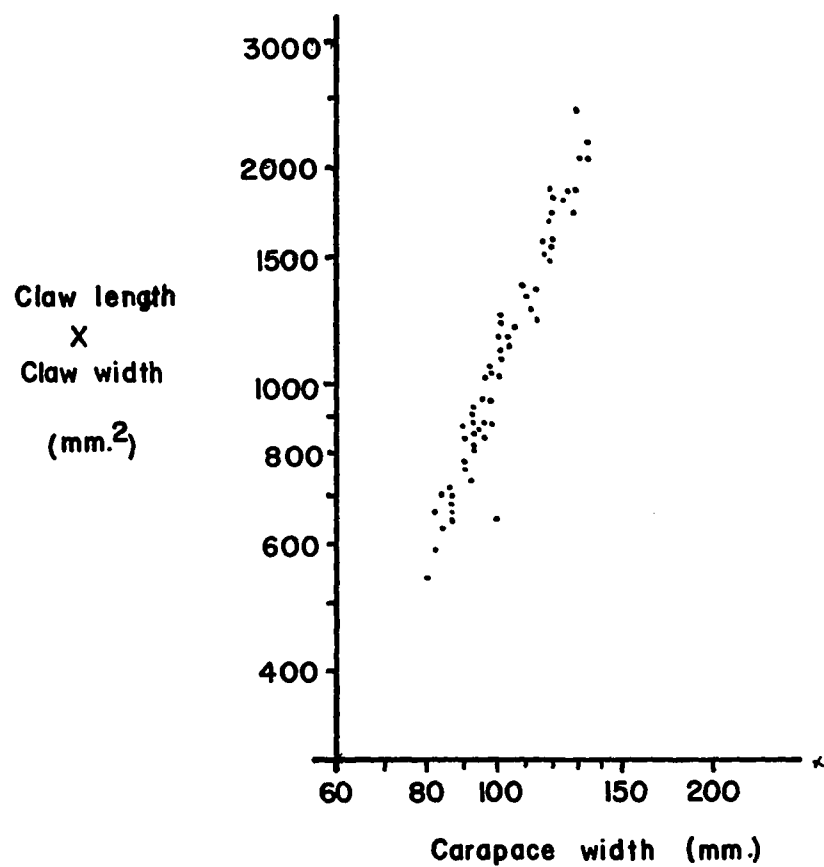


Figure 30. Allometric relationship of claw area to carapace width—males, Cape Breton Trough.

	<u>CHALEUR BAY TROUGH</u>	<u>CAPE BRETON TROUGH</u>
<u>Males</u>		
CW vs. Cl. Area	$\begin{cases} Y_1 = -2.0316 + 2.4271X \\ Y_2 = -2.0439 + 2.5275X \end{cases}$	$Y = -2.5606 + 2.781X$
CW vs. 3rd Abd.	$Y = -.2444 + .8996X$	$Y = -.1491 + .8538X$
<u>Females</u>		
CW vs. Cl. Area		
CW vs. 3rd Abd.	$Y = +.05472 + .8957X$	$Y = .1167 + .9305X$

Table 15. Equations of fitted allometric growth lines for three characters in two areas of the Magdalen Shallows.

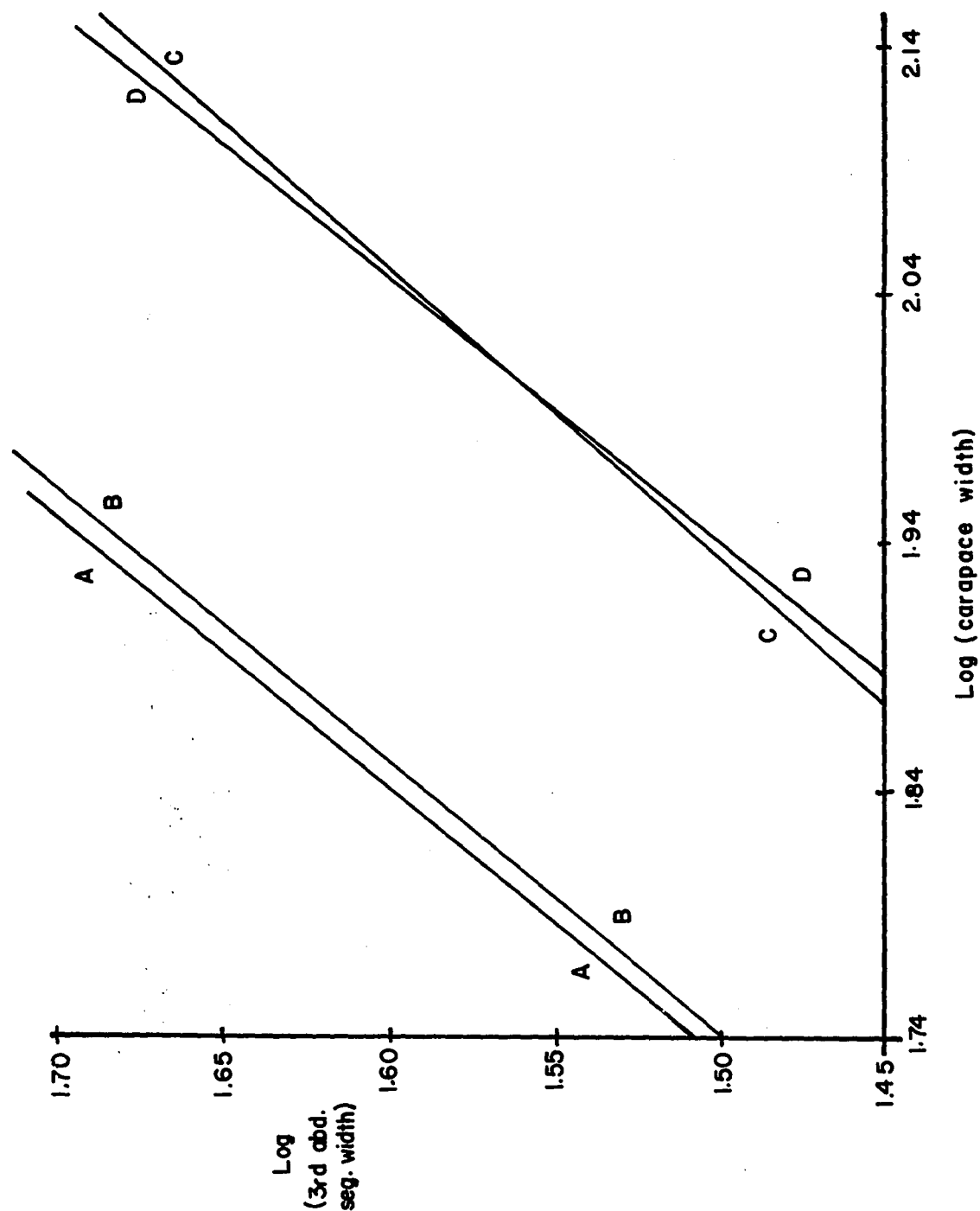


Figure 31. Logarithmic plots of lines fitted to allometric data.

- A. Females, Cape Breton Trough
- B. Females, Chaleur Bay Trough
- C. Males, Cape Breton Trough
- D. Males, Chaleur Bay Trough

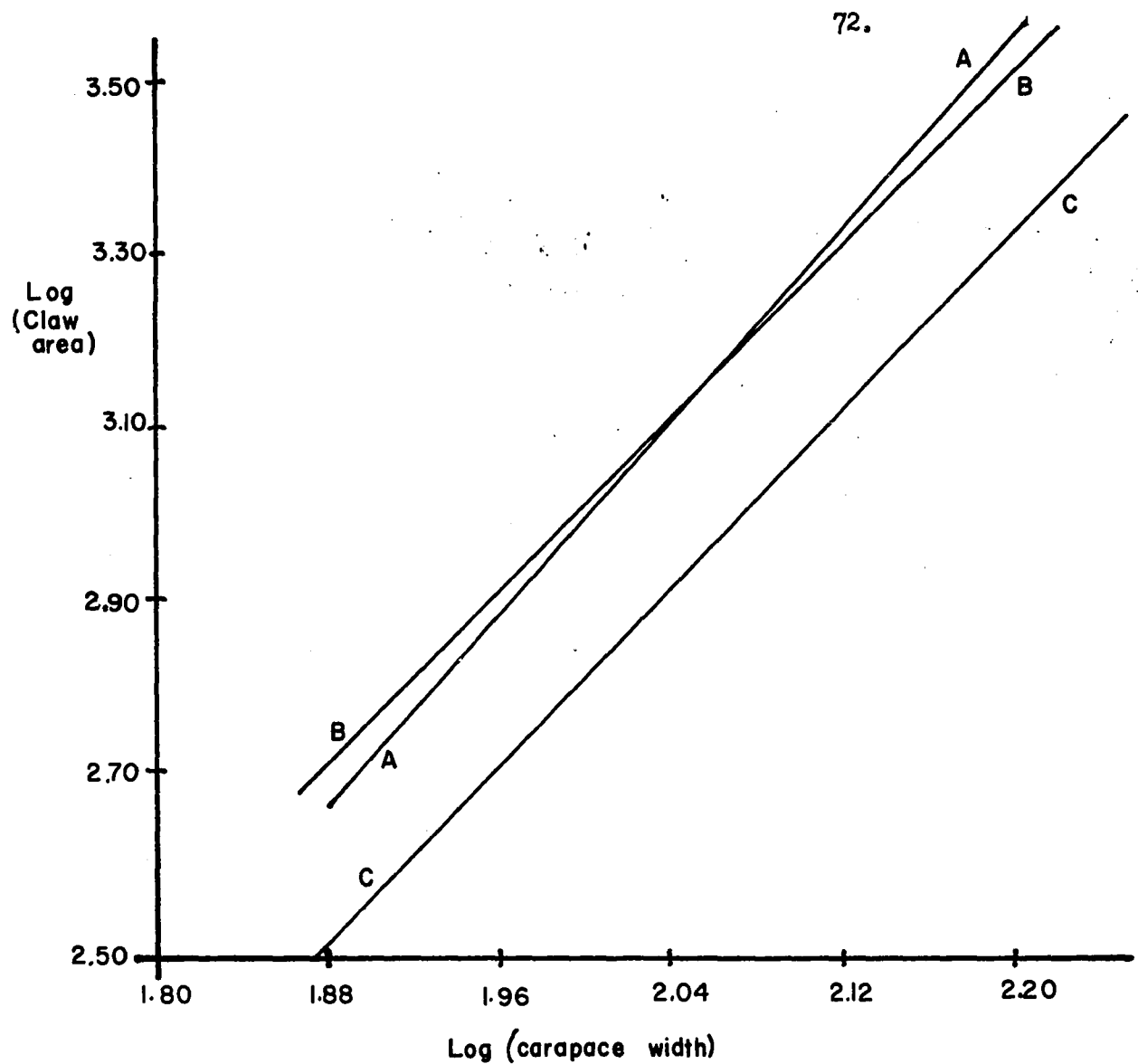


Figure 32. Logarithmic plots of lines fitted to allometric data.

A. Males, Cape Breton Trough

B. Males in final moult class, Chaleur Bay Trough

C. Males, in next-to-final moult class, Chaleur Bay Trough

point of the two lines is 95 mm., which may be considered an average shift point from one line to the next in terms of the population as a whole. The significance of this separation, other than in its marking amount, is not clear--ordinarily one would expect such an abrupt increase in claw size to occur at the moult to maturity, but as will be seen later maturity in male C. opilio occurs much lower in the size range. The separation is, then, simply into two moult groups; the presence of another allometric line below the lower one again is hinted at by the two points at (96 mm., 465mm.) and (98, 430) on the log-log plot (figure 30). It is not clear why two lines did not show up on the plot for males in the Cape Breton Trough, but a second line here is suggested by the points at (94, 680) and (100, 650), (figure 29). Sampling of animals smaller than 100 mm. was not as good with the larger-meshed traps used in the Cape Breton Trough, so males of the lower moult-group were more likely to be excluded than in the Chaleur Bay samples. No separation was noted for males considering abdomen width, nor for females with either claw size or abdomen width; in the case of the females, this was no doubt due to sampling of an insufficient size range. One female of carapace width about 30 mm. was seen with a narrow abdomen; this animal was indistinguishable at a glance from males of the same size or somewhat larger.

Finally allometric lines can be used to show whether or not there is morphological separation of the populations being considered. A problem was encountered in comparing the least-squares fitted lines--the logarithmic values, which cover a small range (1.7 to 2.2), give equations for straight lines for the same characteristic in the two areas which differ considerably in both slope and intercept (see table 15). Visually, over the range under consideration, two lines being compared (for example, for claw area vs. carapace width in males) are very

close, but testing the difference between the two slopes shows them significantly different. A χ^2 test cannot be used, since the Y-values are not probability-generated. It was thought that visual comparison of the various lines would be sufficient, and figures 31 and 32 are used to this end; from these figures it seems apparent that there is no morphological separation of the two C. opilio populations samples in the Magdalen Shallows.

Moulting

Table 16 and figure 33 summarise the proportions of soft-shelled C. opilio males found in samples at different times of year. The combined samples were divided into sizes of less than and greater than 95 mm. carapace width, since the majority of crabs smaller than 95 mm. appear to belong to one moult class and those wider than 95 mm. to another. In animals smaller than 95 mm., moulting activity increases considerably from July to August; after this, no data are available, but Brunel (1962) reported a decrease, in the male population, in soft-shelled crabs of all sizes to September and October, so moulting for the smaller animals is probably at a maximum in August. Moulting into the group of carapace width greater than 95 mm. is spread out over the summer, with no particular maximum indicated. The higher of the two points for November-December is taken from fishery statistics for this period in the Cape Breton Trough, while the lower is from a personal observation in the same area; the latter was for a small catch (of about 60 crabs), and the former is a rough estimate, so the true proportion of newly moulted animals probably lies between these two points. The duration of the soft-shelled stage must be more than two weeks, judging from return of two male crabs tagged while soft-shelled and returned two weeks later still definitely soft.

From these results it appears that male C. opilio remain in the next-to-final moult class for about a year (from August until the following summer). Tag returns suggest that males in the final moult class live for about a year following moulting. Returns in 1967 for animals of carapace width 98 to 124 mm. tagged in August 1966, were as follows: two in June, three in July, three in August, and one in October. The decrease in October, although small, suggests that mortality is occurring; no returns were made after October 1967, although fishing continued through November and part of December. All nine crabs

Time and gear	>95 mm. hard	<95 mm. hard	>95 mm. soft	<95 mm. soft	TOTALS
June 1967 trap	1490 90.7	41 2.5	110 6.7	2 0.1	1643 100.0
July 1966 trawl	68 40.7	56 33.5	41 24.6	2 1.2	167 100.0
1967 trap	753 98.6	11 1.3	1 0.1	- -	765 100.0
August 1967 trawl	649 51.8	152 12.2	369 31.3	55 4.7	1225 100.0
1966 trap	939 82.0	128 11.3	63 5.5	23 1.2	1153 100.0
November- December 1966 Fishery	3204 89.7		368 10.3		3572 100.0

Table 16. Proportions of soft- and hard-shelled animals in catches of male crabs.

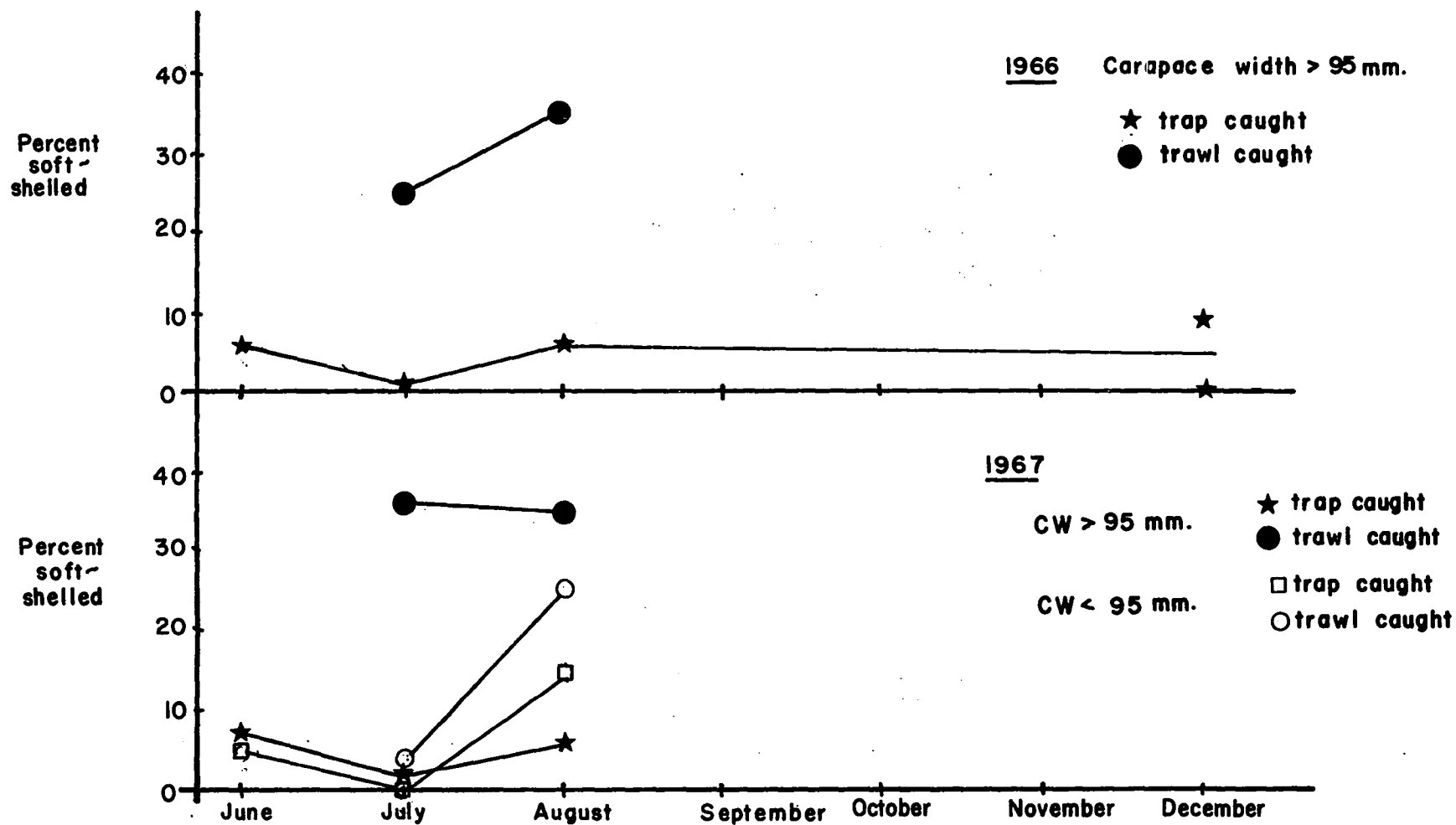


Figure 33. Proportion of soft-shelled male C. opilio in catches during 1966 and 1967.

returned in 1967 had been hard-shelled when tagged, indicating that they had moulted more than two weeks earlier; thus, those returned in August and October had lived for more than a year after the terminal moult.

No soft-shelled females were found in any samples. The significance of this is discussed below.

Gonad states and size at maturity

Female

Figure 34 summarises maturity of females (as judged by presence of eggs) as related to size. Data for the figure are from all samples from the Grande-Riviere area, trap- and trawl-caught, for 1966 and 1967. The 50% maturity point for females is seen to come at 55 mm. carapace width. Sampling inadequacy, however, makes this conclusion somewhat suspect, as below 62 mm. numbers caught decline sharply, and are inadequate to make an exact prediction. Nonetheless, the shape of the curve given here is probably fairly close to that for the whole population, and it seems reasonable to say that 50% maturity for females comes in the 50 mm. to 60 mm. size range. Squires (1965) found 50% maturity of C. opilio females from the Newfoundland-Labrador area to occur at 33 mm. carapace length, equivalent to 40 mm. carapace width. The discrepancy is large enough to be significant, and may be due to our insufficient data for small females.

Eggs found under the abdomens of berried female crabs in June were all bright orange, indicating that they had been recently extruded. In August, the majority of eggs, in the inner portion of the egg mass, were still bright orange, but a few on the periphery of the mass had developed eye spots. These latter remained orange, except for the spots, and were thus not completely developed. Four females caught

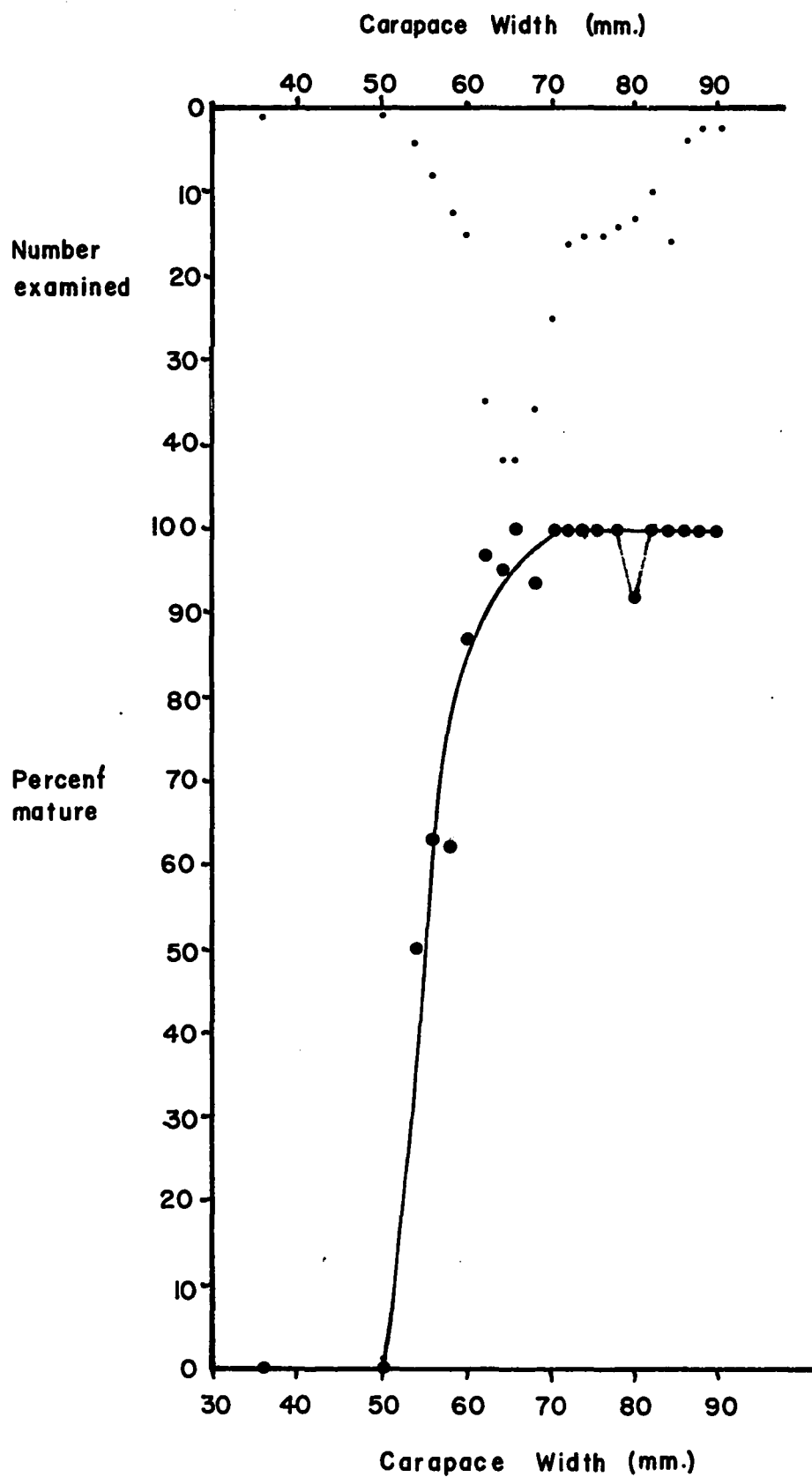


Figure 34. Size-maturity curve, females.

in November and December were also carrying orange eggs, apparently not much further developed than those seen in August. Sampling of females was not possible between December and early May due to ice cover, which was prolonged in 1966; Brunel (1961), however, found females of mature sizes with empty egg-cases on otherwise empty abdomens in May. Egg extrusion, then, must occur in late May and early June, while hatching of larvae must come a short while previous, and females carry their eggs for a year from extrusion to hatching. The larvae are pelagic, and go through a pre-zoea, two zoeae, and a magalops stage (Kurata, 1963), this sequence of stages being typical of the family Majidae (Gurney, 1942).

Berried females taken in August had full, apparently ripe or ripening ovaries, suggesting that breeding would take place following hatching of the eggs. All berried females examined, even in June, were hard-shelled and had a heavy epibiotic growth, indicating that moulting could not have taken place recently. Copulation must, then, take place almost immediately after the larvae are hatched, and without the female's moulting. Such a procedure is unusual for a decapod, in most of which copulation must involve a soft-shelled female. Kato (1956) states that females of the closely-related C. opilio elongatus in the Japan Sea follow this procedure, hatching the larvae in February or March and copulating immediately without moulting.

Fecundity of C. opilio females is apparently high--Brunel (1962) found 30,000 to 80,000 orange eggs per female in June and July, 10,000 to 30,000 fully-developed eggs in May. The difference is striking, and with the presence of eyed eggs among undeveloped ones in August mentioned before, suggests that hatching of the eggs may go on throughout the year, from perhaps August until all are hatched in May. This mechanism is unlikely, however, as one would expect larvae to be released at the optimum time, during the spring plankton bloom.

Male

Four stages of gonad development were noted in male C. opilio preserved in 10% formalin. The first, apparently immature stage, was characterised by the position of the gonads on either side of the stomach, with little differentiation of the various regions--testes at the anterior part of the reproductive structure, vasa deferentia at the posterior end.

The length of the reproductive structures at this stage was some 2 cm. The second stage is thought to represent a mature but undeveloped gonad--the vasa deferentia and testes are differentiated in structure, with the testis a much coiled structure and the vas deferens a long, fairly straight tube with many diverticula branching from it. The testes remain at either side of the stomach, while the vasa deferentia are stretching posteriorly, toward the heart region. Length is increased to 3-4 cm.

The third stage was apparently an intermediate one, with the vasa deferentia beginning to fill with spermatophores; at this point, the vasa deferentia were extending posteriorly toward and around the heart.

The total length of the gonads had increased to about 5 cm., and thickness of the vas deferens mass was of the order of 1.5 cm. This stage was apparently a ripening one, in preparation for copulation, and is referred to in the tables as the "developing" stage. By the fourth stage, that of fully ripened gonads, the vasa deferentia are large and swollen, taking up all available space on both sides of the heart and extending anteriorly almost to the posterior margin of the stomach.

The position of the testes in the latter two stages was hard to determine, probably because of deterioration of tissue in the formalin. In some animals, the testes were seen to lie about midway along the stomach, pressed to each side of it, while in others they were not evident. The

characteristic H-shape of the crustacean gonad pair, with an inter-connection at the posterior margin of the stomach, was not observed in *C. opilio* males; this, too, may have been due to deterioration, although in crabs opened at sea immediately after capture the connection was not observed either. The gonad state and vas deferens thickness of crabs opened at sea corresponded to that described above for preserved animals. The last three states are apparently stages of the mature *C. opilio* gonad, and have been lumped as "mature" in the maturity vs. size table and graph.

Tables 17 and 18, and figure 35, summarise the gonad state of males in relation to size and carapace condition. In this case, carapace condition is described as "old-shell" or "new-shell"; the old-shell designation includes animals with a heavy growth of bryozoan and other epibiotics, while "new-shell" includes animals without much growth. These latter might be called both soft and hard if identified at sea.

From table 17 and figure 35, covering size and gonad state, it will be seen that sampling insufficiency in the lower size ranges has seriously affected results. Except for one animal of 39 mm. carapace width, the smallest examined were in the 60-65 mm. range, and the data for this size give 67% maturity. Further data in the 40-60 mm. range would indeed be desirable, to give an accurate picture of effect of size on maturity; the graph as it stands is an approximation to the actual state of affairs. It is probably true, however, that the 50% maturity point comes somewhere between 45 and 55 mm. carapace width; this is the best generalisation one can make from the graph as it stands. Table 18 shows the effect of carapace condition on degree of development of the mature gonad. Animals 100 mm. and bigger were used for these

Carapace width mm.	<u>Old Shell</u> Number	<u>New Shell</u>	# mature	% mature
60		1	0	0
60		3	2	67
65		1	1	100
70		5	4	80
75		5	4	80
80		2	2	100
85		7	6	86
90		4	4	100
95		3	3	100
100	2	6	8	100
105	2	4	6	100
110		9	9	100
115	4	6	10	100
120	4	4	8	100
125	1	4	5	100
130	2	4	6	100
135	4		4	100

Table 17. Maturity with size of male C. opilio.

<u>Carapace condition</u>	<u>Undeveloped</u>	<u>Developing</u>	<u>Ripe</u>	<u>Total</u>
new	28	8	1	37
	= 93%	= 80%	= 6%	
old	2	2	15	19
	= 7%	= 20%	= 94%	
	30	10	16	

Table 18. Degree of gonad development with carapace condition.
(Male, carapace width ≥ 100).

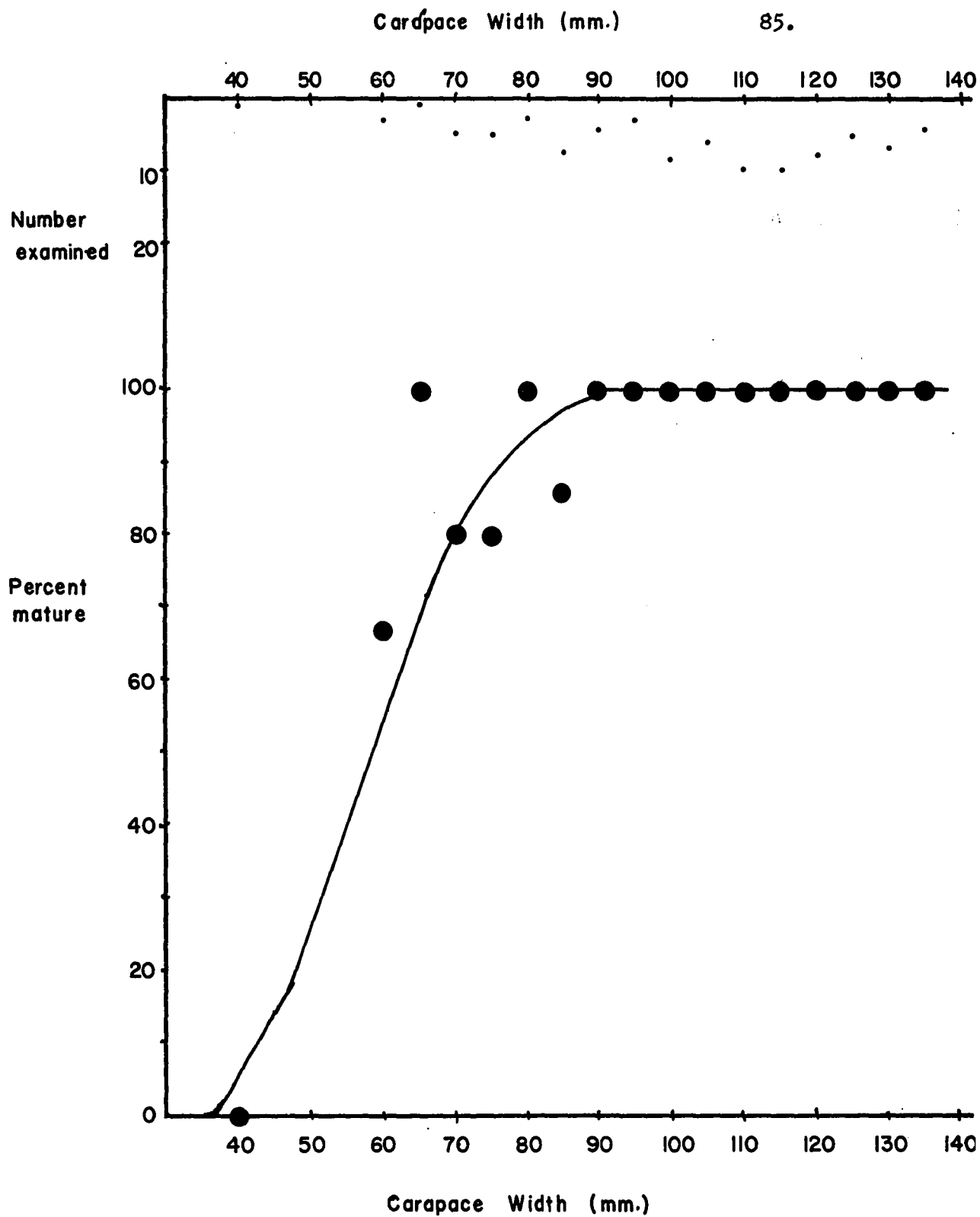


Figure 35. Size-maturity curve, males.

summaries, as all animals below this size were new-shelled and immature. The number of crabs with new and old shells at each size are given in table 17. It is immediately obvious that the great majority of animals with mature but undeveloped gonads are new-shelled, while the majority of animals with ripe gonads are old-shelled.

These results are puzzling. Continuous breeding during the year, or at least during the summer, which might explain a proportion of old-shelled males retaining their reproductive products in August, is argued against by the presence of effectively 100% of mature females carrying immature eggs during the summer, which localises the time of copulation and extrusion of eggs to May and early June. The most reasonable explanation is that the old-shelled ripe males found in the August sample have not copulated in the preceding May and thus retain their full vasa deferentia.

The failure of females to moult after reaching mature sizes, discussed above, explains the difference in maximum size of males and females. Maturity apparently occurs at about the same size in each sex, but after maturity the males undergo several further moults, while the females remain at their size of first maturity. The reason for the females' cessation of growth must lie in their fecundity; production of a large number of eggs must necessarily take up much of the energy input to the females, leaving little for incorporation of other body tissues. This factor would particularly affect a crustacean, in which the moulting process itself takes up energy which might be better used in egg production.

Stomach Contents

Stomach contents of C. opilio were generally ground up thoroughly, leaving the hard parts of the food as the only basis for identification. Soft parts were much digested, usually taking the form of a single mass of tissue, devoid of identifiable features. Hard parts were frequently found in small pieces (ca. 1-3 mm.), so offered little for identification to species except in a few cases. Stomach contents, with their relative frequency, are outlined in table 19. Sedentary polychaetes were found as short sections of empty tubes; one polychaete, only partly digested, was thought to be errant because of the large number of chaetae remaining with the fleshy tissue. Chaetae alone were found in many stomachs, but did not allow categorization as errant or sedentary in origin. Mollusc fragments were the most frequent items in the stomachs; the most frequent type was a simple pelecypod, probably a Yoldia species from hinge teeth attached to some fragments. Three other pelecypod types and one gastropod were also found. Amphipod fragments, chiefly appendages and plates from the body, were quite frequent. Three identifiable head sections were found, all belonging to Paratryphosites abyssi; the fact that all three identifiable amphipods were of this species suggests that it may constitute the bulk of the amphipod diet of C. opilio. Bits of shell with spines and coloration similar to C. opilio itself, found in six stomachs, indicate that the animal is cannibalistic to some extent. The ophiuroid fragments found were small lumps of calcium carbonate, with no particular form, which resembled ground-up ophiuroid samples from the same area both visually and chemically. Fish bones found were chiefly vertebrae and ribs, with one toothed jaw, their sizes suggesting a fish of some 10-15 cm. in length. There was no indication as to whether the fish had been captured while alive and active

<u>Food Item</u>	<u>Number of Stomachs</u>	<u>% of Stomachs</u>
Polychaete:		
Sedentary	20	19
Errant	1	1
Unclassified (chaetae)	45	42
Mollusc:		
<u>Yoldia</u> sp.	80	75
Other pelecypod	11	10
Gastropod	10	9
Crustacean:		
Amphipod	35	33
Benthic decapod (<u>C. opilio</u>)	9	8
Echinoderm:		
Ophiuroid	36	34
Vertebrate		
	13	12
Eggs (unidentified)	11	10
Empty	4	4

Number of stomachs examined: 107

Table 19. Frequency of food items in C. opilio stomachs.

or eaten when already dead; that C. opilio will feed on dead fish is shown by the large trap catches. The source of the eggs found in stomachs is not known, as they do not correlate strongly with any other food item (see table 20). They were 1.8-2.2 mm. in diameter, dark orange to black in color, and were represented only by thin hard cases, with no identifiable contents. Empty stomachs were found in only four animals. Table 20 summarises frequency of occurrence of food item pairs; no strong correlations are indicated, except between items found very frequently, which correlations would be expected if food were randomly selected from available numbers of each type.

Selection of food items cannot be described from these data since the digestion time for each item is not known. The data suggest, however, that C. opilio makes extensive use of the animal food available to it, feeding on a wide variety of groups. The only real example of selection found was for Paratryphosites abyssi of the amphipods in the area surveyed; this is probably due to the abundance of this amphipod, and to its burrowing into the bottom mud at times (P. Brunel, personal communication).

Polychaete	{	Sedentary	Sedentary											
		Errant	1	Errant										
		General	4	0	General									
Mollusc	{	<u>Yoldia</u>	15	1	35	<u>Yoldia</u>								
		Pelecypod	3	0	4	7	Pelecypod							
		Gastropod	1	0	5	9	2	Gastropod						
		Amphipod	7	1	8	30	5	3	Amphipod					
		<u>C. opilio</u>	5	0	2	2	1	0	0	<u>C. opilio</u>				
		Ophiuroid	8	0	16	29	2	2	10	2	Ophiuroid			
		Vertebrate	3	0	4	7	2	0	2	2	1	Vertebrate		
		Egg	4	0	7	9	1	3	5	1	5	0	Egg	
		Empty	0	0	2	8	0	0	2	0	1	2	2	Empty

Number in box indicates % of stomachs in which
item-pair occurred.

Table 20. Frequency of occurrence of pairs of food items
in C. opilio stomachs.

World Distribution and General Discussion

The world distribution of C. opilio is shown in figure 36, sources for this figure being Nesis (1962a), Rathbun (1925), and Squires (1966). It is questionable whether finds in the Seas of Japan and Okhotsk are C. opilio or the subspecies elongatus--certainly the latter is abundant in the Sea of Japan (Kato, 1956)--but the two are very similar and are probably still in the process of differentiation. Figure 37 shows the distribution of subarctic water, as outlined by Dunbar (1968). Subarctic water is defined as water composed of mixed Arctic and non-Arctic elements. In the west Atlantic, the distribution of C. opilio corresponds almost exactly with that of subarctic water, and the crab is also present in the subarctic water between Alaska and Siberia. Waters of the Seas of Japan and Okhotsk, and of the Bering Sea and the northeast Pacific off northern British Columbia, are not subarctic as defined here, but are composed of winter-cooled Pacific waters; indeed, the situation here may be analogous to that in the Gulf of St. Lawrence, where there is very little Arctic influence and the cold intermediate layer is formed in situ in winter. Temperatures at which C. opilio are found in these Pacific areas are somewhat higher than those in the Canadian east--Rathbun (1925) gives a range of -1°C to 10°C , mean temperature being about 4°C , for samples examined by her. The failure of the crab to move into subarctic areas east of Greenland may have been due to the southward currents along east Greenland, preventing dispersal of the larvae, and to water between Greenland and northern Europe too deep to permit the adults to live (Nesis, 1962a).

The exclusion of the crab from Arctic areas must be due to other circumstances than cold alone, since the adults are capable of surviving in Arctic conditions of constant cold in the Magdalen Shallows

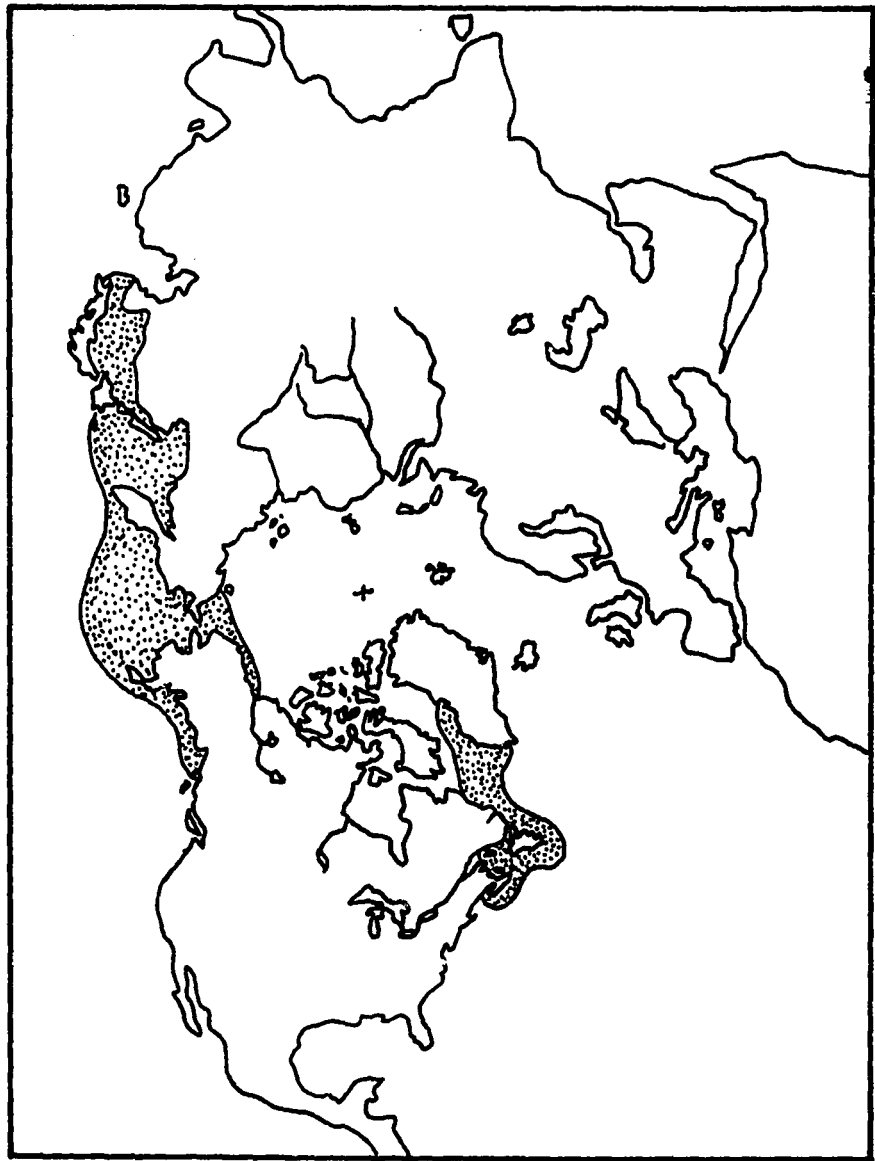


Figure 36. World distribution of Chionoecetes opilio.

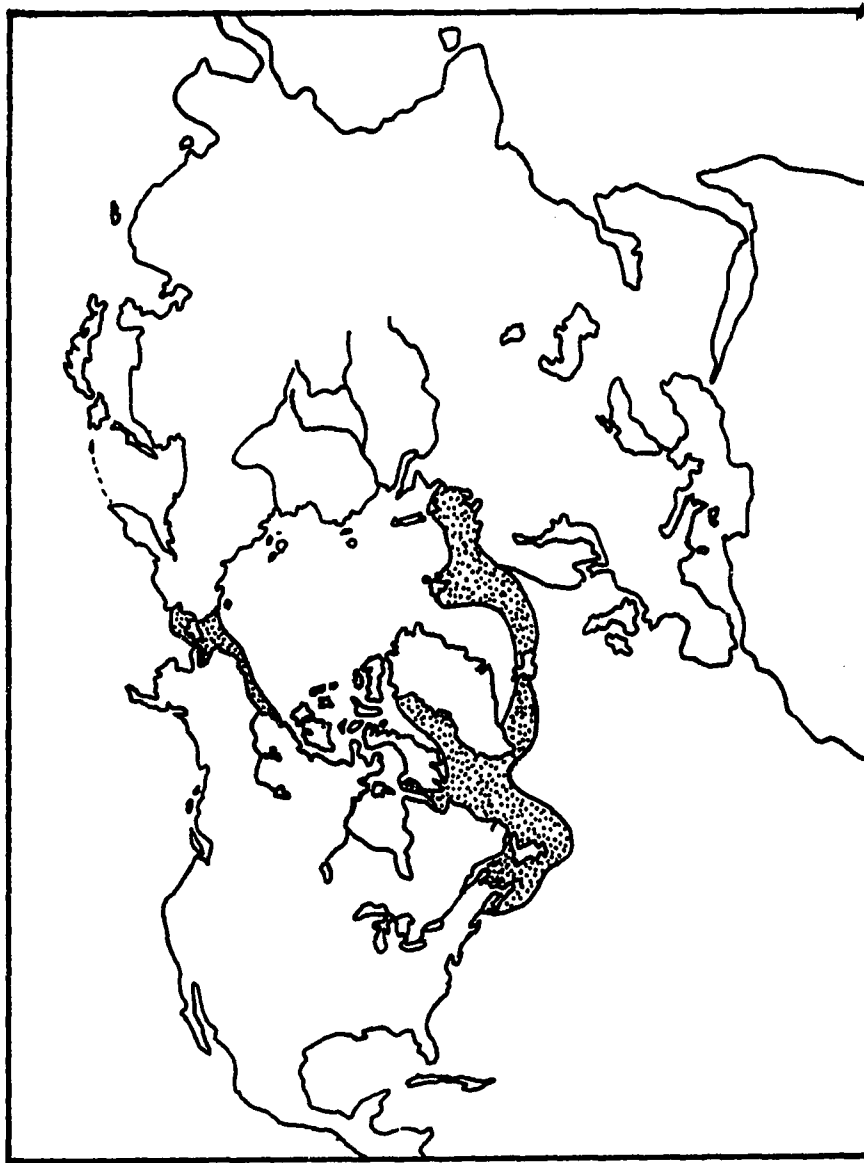


Figure 37. Distribution of subarctic water, after Dunbar (1968).

and elsewhere. Productivity is more probably the limiting factor, particularly in relation to the larvae, which must need a period of bloom to survive in numbers sufficient to maintain the population. Subarctic waters supply both the cold temperatures to which the adults are adapted and the necessary plankton bloom.

The present work suggests factors which might influence the local distribution of the crab within the area of its world distribution. The Magdalen Shallows, with a constant cold bottom environment and highly fluctuating upper water layer, is a typical subarctic (in a broader sense than as used above); environment; within this large area, the distribution of C. opilio is principally determined by bottom sediments, with soft muddy bottoms the influencing factor.

Comparison of the distribution of C. opilio with those of two closely-related brachyuran species (family Majidae, subfamily Pisinae) is instructive. Hyas araneus and Hyas coarctatus have subarctic distributions overlapping that of C. opilio; H. coarctatus is circum-polar, being found in the Barents Sea (Kuznezov, 1964) as well as in the area sketched in figure 1 for C. opilio (same sources as above, for C. opilio), while H. araneus is found in the West Atlantic and in the Barents Sea (same sources). All three species occur in the Magdalen Shallows, and their niches appear to be discontinuous with respect to both sediment type and temperature. H. coarctatus was found at eight stations of the Harengus cruise at temperatures ranging from -0.2°C to 0.8°C . Three of the stations were on mud bottom, and a total of 6 H. coarctatus were taken on these; for the total of 15 stations on mud bottoms, the average catch of this species is 0.4 per tow. The other five stations on sand bottom yielded 7 H. coarctatus, for an average of 0.9 per tow

(8 tows on sand). On the 3 mud stations where H. coarctatus was caught, it was co-existing with C. opilio; on the 5 on sand, C. opilio numbers had decreased, and it is felt by the writer that although the small numbers of H. coarctatus taken do not permit conclusive generalisations to be drawn, H. coarctatus may be more plentiful generally on cold sandy bottoms, which are inhospitable to C. opilio. Hyas araneus was taken at three stations, with bottom temperatures of 0.4°C , to 2.1°C , one on mud bottom, two on sand. In addition, some H. araneus were taken in a series of gear-testing trap sets off eastern Prince Edward Island in June 1966, on sandy bottoms at a depth of 8-14 fathoms; at this depth, summer temperatures would tend to be about 2°C to 5°C . Neither C. opilio nor H. coarctatus was found here. Further evidence of differences in distribution with temperature is given by Squires (1966), who found the three species at these temperature ranges in 80% of finds: C. opilio, -0.5°C to 0.5°C ; H. coarctatus, 0°C to 1°C ; and H. araneus, -1°C to 2.2°C . If Hutchinson's (1957) model of a niche as a hypervolume determined by a number of varying parameters is followed for these three species, temperature range and bottom type could be taken as two parameters in part determining the niche of each species; conditions of each parameter in the Magdalen Shallows, while overlapping to some extent, are sufficiently discontinuous to allow each species to survive. The two Hyas species, in general, are more amenable to life on sandy bottoms than is C. opilio, while the greater temperature range tolerance of H. araneus allows it to live separately from H. coarctatus.

A fourth decapod species, Lithodes maja (Anomura, Paguridae, Lithodidae), has been found (Squires, 1966; F.R.B. unpublished data) in the deep warmer waters of the Gulf of St. Lawrence and of subarctic areas of the northwest Atlantic; it is markedly discontinuous with the three

majid brachyura, 80% of Squires' L. maja being found at temperatures between 3° and 4.5°C.

The preference of C. opilio for mud bottom appears from this study to be explained in terms of retiring behavior. Demersal fish, chiefly cod, and other crabs prey on C. opilio, and soft bottom mud apparently provides a refuge from predators, particularly during daylight hours when "migrations" into the mud appear to be the usual pattern.* Retiring secretive behavior is a part of the life of all brachyura at moulting time; Passano (1960) notes that prior to moulting, crabs seek out the most protected part of their habitat--a burrow for land crabs, an isolated tidal pool or heap of damp algae for intertidal and immediately sublittoral species, or a pile of other crabs in the case of the herbivorous Maja. Brunel (1961) notes that bottom mud provides the safe retreat for C. opilio at its time of moulting.

Nesis (1962a), in a comprehensive discussion of a group of Pacific-West Atlantic subarctic species of which C. opilio is one, indicates the probable course of events leading to dispersal of these animals from Pacific waters to West Greenland. Present day currents are adequate to account for further movement of C. opilio over the West Atlantic; dispersal was probably principally through the pelagic larvae, since the present work suggests that movements of the adults are not great. Unfortunately, the length of the larval period is not known; this would be valuable principally in estimating speed of dispersal and the relation between

* Burying is probably particularly advantageous to the females, who have sacrificed the potential anti-predation strength of large size for increased egg production.

various populations at present. The Labrador current must account for spread of C. opilio over the continental shelf off Labrador and Newfoundland, including the banks off Newfoundland. Two branches of the Labrador current influence circulation in the Gulf of St. Lawrence (Hachey, 1961; Bumpus and Lauzier, 1965); one, of quite limited influence, southward through the Strait of Belle Isle, would allow distribution of the crab into the northern Gulf, while the other, along the south shore of Newfoundland and north shore of the Gulf, would permit dispersal into these areas and to Anticosti Island. The Gaspé current, from the St. Lawrence estuary southward off the Gaspé coast, is sufficient for the establishment of the Gaspé population; the Cape Breton population may have been settled from the Gaspé via the surface drift eastward across the Magdalen Shallows, or directly from the inflow through Cabot Strait. The former route is the less likely; drift is slow in the spring (Bumpus and Lauzier (1965) give a maximum speed of 3-5 miles per day) when the larvae are in the plankton, and movement over a period of years from west to east is unlikely, since conditions in the central area of the Shallows do not permit significant numbers of the animal to live. If, however, larval life goes on for over a month, larvae would be able to drift the 150-odd miles from the west end to the east end of the Shallows at the given current speed. Actual contact between populations is not, however, necessary to explain the morphological equivalence of C. opilio from Chaleur Bay and Cape Breton troughs; environmental conditions in the two areas are apparently so similar that no differential growth of the body parts would be expected.

Adaptations to the cold productive environment are most apparent in the reproductive life of C. opilio. Under increasingly cold conditions, larval forms of benthic animals are increasingly suppressed

(Thorson, 1950), but in the case of this animal the advantages at the ultimate level are apparently greater with pelagic larvae than without. The chief advantage in retaining pelagic larvae may well be greater possibilities of dispersion; indeed it is through the larvae that dispersal from Pacific to Atlantic probably took place, and, as noted above, it is apparently the lack of conditions suitable to pelagic larvae that prohibits life of C. opilio in the Arctic. Given an animal which retains pelagic larvae, then, sufficient numbers of larvae must be hatched to ensure maintenance of population; apparently very large numbers hatched are required of C. opilio, since female fecundity is high. So many eggs are produced by the individual female that moulting and growth are suppressed, so that all the available energy input can be put into reproductive products. This phenomenon has not, to the writer's knowledge, been described for other brachyuran species, but other drains on the energy resources of brachyura can inhibit moulting to varying extents. Passano (1960) notes that starvation usually inhibits moulting (although in a few species it increases moult frequency), that the intermoult period is lengthened in many species during the period of gonadal maturation, and that parasitism by the rhizocephalan Sacculina slows and finally halts moulting in some crabs. Thus it seems reasonable that the long-term demand on the female for many eggs overrides any advantage that might come from sizes as large as the males.

The importance of the reproductive products is emphasized by the amount of time spent in their development; developing ovaries and ripening vasa deferentia were found in August, long before copulation and egg extrusion in the following spring. Development of the eggs is slow, taking almost a year from extrusion to hatching. Classically, this might be explained as a direct effect of temperature on growth rate,

but this principle is questionable. A better explanation of the phenomenon lies in the effect of growth rate on growth efficiency; a slow chemical reaction is more efficient, i.e. produces less entropy, than a fast one, and it would be expected that nutrient stored in an egg could be incorporated into complex structures more efficiently over a long period of time than over a brief one. This principle is particularly applicable to an animal such as C. opilio where eggs are numerous and the amount of yolk in each is small.

Growth per moult in large adult male C. opilio is not low, compared to other decapod crustacea. Growth rates of several species, for approximately equivalent sizes, are given in table 21. Unfortunately weight increments for Cancer magister and Callinectes sapidus are not available. Growth in this moult of C. opilio is fairly high, compared with the other species; it is comparable to that of female Callinectes sapidus, the commercial crab of Maryland and a warm water species. P. camschatica, the Alaska king crab, is a subarctic sublittoral species in the Pacific and Bering Sea, while H. americanus (the American lobster) and C. magister (the Pacific edible crab) occupy the shallow sublittoral in low subarctic and boreal Atlantic and Pacific waters; growth per moult in each is lower than for C. opilio. Apparently other factors than cold alone are acting to control growth in these species. Growth rates for other majid crabs have not been studied, so an intrafamilial comparison cannot be made.

Species	Mean size of moult class mm.	Size Increment mm.	% increment	Weight increment %
<u>Homarus</u> <u>americanus</u> ¹	150-200		12-15	43-54
<u>Paralithodes</u> <u>camschatica</u> ²	100-170	16 (constant over size range)	9.4-16.0	not given
<u>Cancer</u> <u>magister</u> ³	80.5 95.8 112.6 130.0	15.3 16.8 17.4	19.0 17.6 15.4	not given
<u>Callinectes</u> <u>sapidus</u> ⁴				
males	85.4 106.4 133.2	21.0 26.8	24.6 25.2	not given
females	86.9 116.7 161.4	29.8 44.7	34.3 38.3	not given
<u>Chionoecetes</u> <u>opilio</u>				
males	79 111	31	38	140

¹ Wilder (1953)² Weber and Miyahara (1962)³ Mackay and Weymouth (1935)⁴ Newcombe (1948)

Table 21. Growth per moult in some large decapods.

Summary

Samples of Chionoecetes opilio and environmental data were collected in the summers of 1966 and 1967 in the Chaleur Bay, Shediac, and Cape Breton troughs of the Magdalen Shallows. Locations of C. opilio catches in groundfish surveys between 1962 and 1966 throughout the Magdalen Shallows were provided by the Atlantic Biological Station of the Fisheries Research Board.

1. Presence of C. opilio was not found to be correlated with depth; it was, however, found at a lower mean temperature than that for all stations, being most common at temperatures of -1°C to 2°C . The crab was significantly more abundant on predominantly mud bottoms than on sand bottoms. These conclusions are emphasized by the distribution of C. opilio in the Magdalen Shallows; it is most commonly found in trough areas whose bottoms are predominantly mud and whose depths of 40-90 fathoms give year-round temperatures between -1°C to 2°C .

2. Night trawl catches of C. opilio are about five times greater than day catches. This is suggested to be due to daytime burying in the bottom mud, probably for predator avoidance.

3. Both traps and otter trawls appear to select for large C. opilio.

4. Two size frequency distributions for male C. opilio were analysed using probability paper. Growth in the final moult for males was estimated as 31 mm. (38%) carapace width, and, using a weight vs. carapace width curve, as 350 gm. (140% by weight).

5. Allometric graphs of two characters vs. carapace width were used to show that there is no morphometric difference between populations of C. opilio from the Chaleur Bay and Cape Breton troughs. The final moult for male crabs in the Chaleur Bay trough was demonstrated;

position of this moult was in good agreement with the results of the size frequency distribution analysis.

6. Moulting of male crabs of carapace width less than 95 mm. is apparently concentrated in August, while for males larger than 95 mm. moulting occurs throughout the summer. Mature females found were all old-shelled, indicating that moulting had not taken place before copulation and extrusion of eggs.

7. The 50% maturity points for females and for males come at approximately the same position in the size range, between 45 and 60 mm. carapace width. This is not a particularly good estimate, due to the small number of crabs of low carapace width in the samples.

8. Extrusion of eggs occurs in late May and early June; eggs are hatched almost a year later, in late April and early May. Developing ovaries in ovigerous females suggests that females breed more than once.

9. Stomach contents of 135 crabs are tabulated. No correlation of pairs of food items was found, and there is no indication of selection of food items from those available.

10. World distribution of C. opilio is summarised from several sources. It is subarctic in the northwest Atlantic and the Bering Straits area, and is found in winter-cooled shelf waters of the Seas of Japan and Okhotsk, Bering Sea, and northeast Pacific. Aspects of the crab's life studied in this paper are discussed in relation to its world and local distributions.

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